

IC3 GLOBAL STANDARD 6

LEVEL 1

TECHNOLOGY BASICS

TEXNOLOGIYA ASOSLARI

Level 1

Fundamental Concepts and Essential Components



Lesson 1: Technology Basics

Lesson Objectives

In this lesson you will explore the following concepts:

- ☐ 1.1 Access and navigate between digital environments
- ☐ 1.2 Identify digital devices and connections
- ☐ 1.3 Explain fundamental software concepts
- ☐ 1.4 Explain fundamental hardware concepts
- ☐ 1.5 Explain fundamental operating system concepts
- ☐ 1.6 Explain fundamental networking concepts

English — Uzbek | Kengaytirilgan o‘quv nashri

Kitob tuzilishi / Book structure

- Original source page — manbadagi sahifa ko‘rinishi.
- English — manba matni va asosiy terminlar.
- O‘zbekcha — o‘quvchi uchun tushunarli tarjima/talqin.
- Key Terms — muhim IC3 terminlari.
- Quick Check — mavzuni mustahkamlash uchun qisqa nazorat.
- Study Note — imtihonda e’tibor beriladigan asosiy nuqtalar.

Asosiy terminologiya / Core terminology

English term	O‘zbekcha
Operating system (OS)	Operatsion tizim
Hardware	Apparat ta’minoti / qurilmalarning jismoniy qismi
Software	Dasturiy ta’minot
Application program	Amaliy dastur
Input device	Kiritish qurilmasi
Output device	Chiqarish qurilmasi
Peripheral device	Tashqi/periferik qurilma
Network	Tarmoq
Internet	Internet
Browser	Brauzer
Download	Yuklab olish
Upload	Yuklash / yuborish
Bluetooth	Bluetooth simsiz aloqa texnologiyasi
Cloud-based software	Bulutga asoslangan dasturiy ta’minot
Open-source	Ochiq kodli
Proprietary	Xususiy / yopiq litsenziyali

Lesson Objectives / Dars maqsadlari

Objective	English	O'zbekcha
1.1	Access and navigate between digital environments	Raqamli muhitlarga kirish va ular orasida navigatsiya qilish
1.2	Identify digital devices and connections	Raqamli qurilmalar va ulanishlarni aniqlash
1.3	Explain fundamental software concepts	Dasturiy ta'minotning asosiy tushunchalarini tushuntirish
1.4	Explain fundamental hardware concepts	Apparat ta'minotining asosiy tushunchalarini tushuntirish
1.5	Explain fundamental operating system concepts	Operatsion tizimning asosiy tushunchalarini tushuntirish
1.6	Explain fundamental networking concepts	Tarmoqning asosiy tushunchalarini tushuntirish

LESSON 1 — TECHNOLOGY BASICS | SECTION 1

English — O‘zbekcha

ENGLISH	O‘ZBEK
ENGLISH	O‘ZBEK
IC3 Global Standard 6 Series Level 1 Fundamental Concepts and Essential Components Lesson 1: Technology Basics	IC3 Global Standard 6 seriyasi 1-daraja Asosiy tushunchalar va muhim komponentlar 1-dars: Texnologiya asoslari
Lesson Objectives	Dars maqsadlari
In this lesson you will explore the following concepts:	Ushbu darsda quyidagi tushunchalarni o‘rganasiz:
1.1 Access and navigate between digital environments	1.1 Raqamli muhitlarga kirish va ular orasida navigatsiya qilish
1.2 Identify digital devices and connections	1.2 Raqamli qurilmalar va ulanishlarni aniqlash
1.3 Explain fundamental software concepts	1.3 Dasturiy ta’minotning asosiy tushunchalarini tushuntirish
1.4 Explain fundamental hardware concepts	1.4 Apparat ta’minotining asosiy tushunchalarini tushuntirish
1.5 Explain fundamental operating system concepts	1.5 Operatsion tizimning asosiy tushunchalarini tushuntirish
1.6 Explain fundamental networking concepts	1.6 Tarmoq texnologiyalarining asosiy tushunchalarini tushuntirish

Study Note / O‘quv eslatmasi

- Hardware — Apparat ta’minoti / qurilmalarning jismoniy qismi
- Software — Dasturiy ta’minot
- Network — Tarmoq

Quick Check / Tezkor nazorat

1. What is the main concept explained in this section? / Ushbu bo‘limning asosiy tushunchasi nima?
2. Which English technical terms should be remembered? / Qaysi texnik English terminlarini yodlash kerak?
3. Can you explain the concept in Uzbek without looking at the text? / Matnga qaramasdan tushunchani o‘zbekcha izohlay olasizmi?

LESSON 1 — TECHNOLOGY BASICS | SECTION 2

Manba sahifasi / Source page: 2

Lesson 1: Technology Basics

IC3 Global Standard 6 – Level 1

What is an Operating System?

Objective 1.1.1

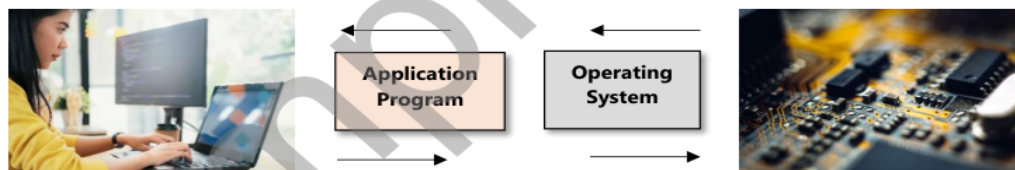
A computer is essentially a collection of electrical and mechanical parts referred to as *hardware*. Each hardware part performs its own function, and in order for all the parts to work together, they must be able to communicate. Instead of communicating directly with one another, the various components communicate via an operating system.

- The *operating system* (or OS) is a computer program.
- A *program* is a sequence of instructions that guides the computer through the performance of a specific task or sequence of tasks.
- The individual lines of instruction are referred to as *code*.
- The term *software* refers to any program that makes a computer run – including operating systems and application programs.

Every computer requires an operating system in order to function. Think of an operating system as the central manager of communication, coordination, and control.

Application programs, on the other hand, allow you to be productive – to create documents or complete specific tasks such as creating a budget, removing red-eye from a photograph, searching the Internet or scheduling a meeting.

Application programs use the operating system to control the hardware functions of the computer, and you, as a computer user, interact with the operating system whenever you log on to the computer, or type on the keyboard, tap on a touchscreen, or use an application program such as an email program or a word processor.



It works like this:

As you work in an application program, the program interacts with the operating system behind the scenes.

The operating system:

- manages hardware devices, and
- controls communication among hardware devices, and
- controls communication between application programs and hardware devices, and
- manages files stored on the computer.

English — O‘zbekcha

ENGLISH	O‘ZBEK
ENGLISH	O‘ZBEK
What is an Operating System? Objective 1.1.1	Operatsion tizim nima? Maqsad 1.1.1
A computer is essentially a collection of electrical and mechanical parts referred to as hardware. Each hardware part performs its own function, and in order for all the parts to work together, they must be able to communicate. Instead of communicating directly with one another, the various components communicate via an operating system.	Kompyuter, mohiyatan, apparat ta’minoti (hardware) deb ataladigan elektr va mexanik qismlar majmuasidir. Har bir apparat qismi o‘z vazifasini bajaradi va barcha qismlar birgalikda ishlashi uchun ular o‘zaro aloqa qila olishi kerak. Turli komponentlar bir-biri bilan bevosita emas, balki operatsion tizim orqali muloqot qiladi.
- The operating system (or OS) is a computer program. - A program is a sequence of instructions that guides the computer through the performance of a specific task or sequence of tasks. - The individual lines of instruction are referred to as code. - The term software refers to any program that makes a computer run – including operating systems and application programs.	- Operatsion tizim (yoki OS) — kompyuter dasturidir. - Dastur — kompyuterni muayyan vazifa yoki vazifalar ketma-ketligini bajarishga yo‘naltiruvchi ko‘rsatmalar ketma-ketligidir. - Ko‘rsatmalarning alohida satrlari kod deb ataladi. - Dasturiy ta’minot (software) atamasi kompyuterning ishlashini ta’minlaydigan har qanday dasturga, jumladan operatsion tizimlar va amaliy dasturlarga nisbatan qo‘llanadi.
Every computer requires an operating system in order to function. Think of an operating system as the central manager of communication, coordination, and control.	Har bir kompyuter ishlashi uchun operatsion tizimga muhtoj. Operatsion tizimni aloqa, muvofiqlashtirish va boshqaruvning markaziy menejeri sifatida tasavvur qilish mumkin.
Application programs, on the other hand, allow you to be productive – to create documents or complete specific tasks such as creating a budget, removing red-eye from a photograph, searching the Internet or scheduling a meeting.	Amaliy dasturlar esa foydalanuvchiga samarali ishlash imkonini beradi: hujjatlar yaratish, budjet tuzish, fotosuratdagi qizil ko‘z effektini olib tashlash, Internetdan qidirish yoki uchrashuvni rejalashtirish kabi aniq vazifalarni bajarishga yordam beradi.
Application programs use the operating system to control the hardware functions of the computer, and you, as a computer user, interact with the operating system whenever you log on to the computer, or type on the keyboard, tap on a touchscreen, or use an application program such as an email program or a word processor.	Amaliy dasturlar kompyuter apparatining funksiyalarini boshqarish uchun operatsion tizimdan foydalanadi. Siz esa kompyuter foydalanuvchisi sifatida tizimga kirganingizda, klaviaturada yozganingizda, sensorli ekranga teginganingizda yoki elektron pochta dasturi yoxud matn protsessori kabi amaliy dasturdan foydalanganingizda operatsion tizim bilan o‘zaro ishlaysiz.
Application Program → Operating System	Amaliy dastur → Operatsion tizim
It works like this: As you work in an application program, the program interacts with the operating system behind the scenes. The operating system: - manages hardware devices; - controls communication among hardware devices; - controls communication between application programs and hardware devices; - manages files stored on the computer.	Bu jarayon quyidagicha ishlaydi: Amaliy dasturda ishlayotganingizda, dastur fon rejimida operatsion tizim bilan o‘zaro aloqada bo‘ladi. Operatsion tizim: - apparat qurilmalarini boshqaradi; - apparat qurilmalari o‘rtasidagi aloqani nazorat qiladi; - amaliy dasturlar va apparat qurilmalari o‘rtasidagi aloqani nazorat qiladi; - kompyuterda saqlangan fayllarni boshqaradi.

Study Note / O‘quv eslatmasi

- Hardware — Apparat ta’minoti / qurilmalarning jismoniy qismi
- Software — Dasturiy ta’minot
- Application program — Amaliy dastur
- Internet — Internet

Quick Check / Tezkor nazorat

4. What is the main concept explained in this section? / Ushbu bo‘limning asosiy tushunchasi nima?
5. Which English technical terms should be remembered? / Qaysi texnik English terminlarini yodlash kerak?
6. Can you explain the concept in Uzbek without looking at the text? / Matnga qaramasdan tushunchani o‘zbekcha izohlay olasizmi?

LESSON 1 — TECHNOLOGY BASICS | SECTION 3

Manba sahifasi / Source page: 3

IC3 Global Standard 6 – Level 1

Lesson 1: Technology Basics

Desktop Operating Systems

There are many types of operating systems in wide use today because there are many types of devices. In this course, you will focus on desktop operating systems and mobile operating systems.

A desktop operating system is used on desktop and laptop computers. A desktop operating system is a large and robust program. It offers many features, and provides a great deal of power and control over how the computer runs.

- **Operating system version** – refers to the specific code-base used to develop the operating system, such as Windows 7 and Windows 10 are different versions of the Windows operating system.
- **Operating system edition** – determines which features are available, such as Windows Home, Professional, or Enterprise.

Today, you can choose from several operating systems. You might like Windows, while your friend or classmate prefers Mac OS X, and your teacher or colleague likes Linux.

Almost all operating systems include a *Desktop* which is the screen that opens once you log on and are ready to begin using the device. Operating systems usually include colorful backgrounds, icons or buttons for accessing commands, and a status bar to show which programs are open and running. The Desktop is the central place from which a user interacts with the computer.

Mobile Operating Systems

These operating systems are used on smart phones, and other handheld (or mobile) devices; the options for each system vary depending on the type of handheld device. Popular handheld operating systems include Android, iOS, and Blackberry.

Embedded operating systems manage and control operations on the specific type of equipment for which they are designed, such as a vehicle, a machine that controls robotic manufacturing, or a piece of medical equipment. When the equipment is turned on, the embedded operating system loads into memory. Embedded operating systems are designed to be compact and are highly specialized; they include only the functions that are required by the specific devices for which they are developed.

Case Study 1

An operating system is the most important software on a computer. It is often referred to as “the brain” of the computer. All computers must have an operating system.

Which of the following options is NOT a responsibility of the operating system?

- Providing access to the Internet and web sites.
- Managing the activity between the software and hardware on the computer.
- Providing a user interface to communicate with the computer.
- Managing access to programs and data through user authentication.

English — O‘zbekcha

ENGLISH	O‘ZBEK
ENGLISH	O‘ZBEK
Desktop Operating Systems	Stol kompyuterlari uchun operatsion tizimlar
There are many types of operating systems in wide use today because there are many types of devices. In this course, you will focus on desktop operating systems and mobile operating systems.	Bugungi kunda qurilmalar turlari ko‘p bo‘lgani sababli operatsion tizimlarning ham ko‘plab turlari keng qo‘llanadi. Ushbu kursda stol kompyuterlari va mobil qurilmalar uchun operatsion tizimlarga e‘tibor qaratiladi.
A desktop operating system is used on desktop and laptop computers. A desktop operating system is a large and robust program. It offers many features, and provides a great deal of power and control over how the computer runs.	Stol kompyuterlari uchun operatsion tizim stol kompyuterlari va noutbuklarda ishlatiladi. U katta va barqaror dastur bo‘lib, ko‘plab funksiyalarni taklif etadi hamda kompyuterning qanday ishlashini boshqarish va nazorat qilish uchun keng imkoniyat beradi.
- Operating system version – refers to the specific code-base used to develop the operating system, such as Windows 7 and Windows 10 are different versions of the Windows operating system. - Operating system edition – determines which features are available, such as Windows Home, Professional, or Enterprise.	- Operatsion tizim versiyasi — operatsion tizimni ishlab chiqishda foydalanilgan muayyan kod bazasini anglatadi. Masalan, Windows 7 va Windows 10 — Windows operatsion tizimining turli versiyalaridir. - Operatsion tizim nashri (edition) — qaysi funksiyalar mavjudligini belgilaydi; masalan, Windows Home, Professional yoki Enterprise.
Today, you can choose from several operating systems. You might like Windows, while your friend or classmate prefers Mac OS X, and your teacher or colleague likes Linux.	Bugungi kunda siz bir nechta operatsion tizimlardan birini tanlashingiz mumkin. Siz Windows’ni yoqtirishingiz mumkin, do‘stingiz yoki sinfdoshingiz Mac OS X’ni, o‘qituvchingiz yoki hamkasbingiz esa Linux’ni afzal ko‘rishi mumkin.
Almost all operating systems include a Desktop which is the screen that opens once you log on and are ready to begin using the device. Operating systems usually include colorful backgrounds, icons or buttons for accessing commands, and a status bar to show which programs are open and running. The Desktop is the central place from which a user interacts with the computer.	Deyarli barcha operatsion tizimlarda Desktop mavjud. Bu foydalanuvchi tizimga kirgach, qurilmadan foydalanishni boshlash uchun ochiladigan asosiy ekran hisoblanadi. Operatsion tizimlarda odatda rangli fon, buyruqlarga kirish uchun ikonka yoki tugmalar hamda qaysi dasturlar ochiq va ishlayotganini ko‘rsatuvchi holat satri bo‘ladi. Desktop foydalanuvchining kompyuter bilan o‘zaro ishlashining markaziy joyidir.
Mobile Operating Systems	Mobil operatsion tizimlar
These operating systems are used on smart phones, and other handheld (or mobile) devices; the options for each system vary depending on the type of handheld device. Popular handheld operating systems include Android, iOS, and Blackberry.	Bu operatsion tizimlar smartfonlar va boshqa qo‘lda olib yuriladigan (mobil) qurilmalarda ishlatiladi; har bir tizimning imkoniyatlari qurilma turiga qarab farq qiladi. Keng tarqalgan mobil operatsion tizimlarga Android, iOS va BlackBerry kiradi.
Embedded operating systems manage and control operations on the specific type of equipment for which they are designed, such as a vehicle, a machine that controls robotic manufacturing, or a piece of medical equipment. When the equipment is turned on, the embedded operating system loads into memory.	O‘rnatilgan (embedded) operatsion tizimlar ular uchun maxsus ishlab chiqilgan muayyan uskunalarning ishlashini boshqaradi va nazorat qiladi. Masalan, transport vositasi, robotlashtirilgan ishlab chiqarishni boshqaruvchi mashina yoki tibbiy uskuna. Uskuna yoqilganda, o‘rnatilgan operatsion tizim xotiraga yuklanadi.
Embedded operating systems are designed to be compact and are highly specialized; they include only the functions that are required by the specific devices for which they are developed.	O‘rnatilgan operatsion tizimlar ixcham va yuqori darajada ixtisoslashgan bo‘lishi uchun yaratiladi; ular faqat ishlab chiqilgan muayyan qurilma uchun zarur funksiyalarni o‘z ichiga oladi.
Case Study 1 An operating system is the most important software on a computer. It is often referred to as “the brain” of the computer. All computers must have an operating system. Which of the following options is NOT a responsibility of the operating system? a. Providing access to the Internet and web sites. b. Managing the activity between the software and hardware on the computer. c. Providing a user interface to communicate with the computer. d. Managing access to programs and data through user authentication.	1-amaliy holat Operatsion tizim kompyuterdagi eng muhim dasturiy ta‘minot hisoblanadi. U ko‘pincha kompyuterning “miyasi” deb ataladi. Barcha kompyuterlarda operatsion tizim bo‘lishi kerak. Quyidagi variantlardan qaysi biri operatsion tizimning vazifasi EMAS? a. Internet va veb-saytlarga kirishni ta‘minlash. b. Kompyuterdagi dasturiy va apparat ta‘minot o‘rtasidagi faoliyatni boshqarish. c. Kompyuter bilan muloqot qilish uchun foydalanuvchi interfeysini ta‘minlash. d. Foydalanuvchini autentifikatsiya qilish orqali dasturlar va ma‘lumotlarga kirishni boshqarish.

Study Note / O‘quv eslatmasi

- Hardware — Apparat ta‘minoti / qurilmalarning jismoniy qismi
- Software — Dasturiy ta‘minot

- Internet — Internet

Quick Check / Tezkor nazorat

7. What is the main concept explained in this section? / Ushbu bo‘limning asosiy tushunchasi nima?
8. Which English technical terms should be remembered? / Qaysi texnik English terminlarini yodlash kerak?
9. Can you explain the concept in Uzbek without looking at the text? / Matnga qaramasdan tushunchani o‘zbekcha izohlay olasizmi?

LESSON 1 — TECHNOLOGY BASICS | SECTION 4

Manba sahifasi / Source page: 4

Lesson 1: Technology Basics

IC3 Global Standard 6 – Level 1

Meet the Web Browser

Objective 1.1.2

Web browsers (or simply, browsers) are software applications that enable users to easily access, view and navigate web pages on the Internet. Browsers allow you to interact with websites and to experience the amazing rich media that is available on the World Wide Web.

You may be familiar with several browsers, such as Microsoft Edge, Mozilla Firefox, Google Chrome, Apple Safari and Opera.

While a browser's primary function is to retrieve pages from a web server and display them on your screen, you should understand that browsers are complex software applications that provide a number of services. They can be configured to suit your working style and keep you safe while you are online.

Using the Address Bar

To visit a web page, you enter its URL into the browser address bar and press ENTER. The browser requests the page from the web server, and when it is received, the browser displays the page within the browser window. The address bar displays the URL of the page currently displayed in the browser window. If the webpage includes a title, it is displayed in the window tab.



You can visit any web page by typing its URL into the address bar and pressing ENTER.

As you navigate to other pages within a website (or as you visit pages on other websites), the URL shown in the address field updates to show the address of the current page. Most browsers maintain a history of URLs that you can access from the address bar. Clicking a URL displayed in the address bar history list has the same effect as entering the URL directly into the address bar.

In addition to displaying the URL, the address bar often includes several buttons. The Microsoft Edge address bar is shown in the previous screen.

Browser Scroll Bars

Browser windows (like other windows you may be used to seeing on a computer screen) support horizontal and vertical scroll bars that display when a window is too small to display all the contents of a web page. The following figure shows a browser window with a vertical scroll bar.

English — O‘zbekcha

ENGLISH	O‘ZBEK
ENGLISH	O‘ZBEK
Meet the Web Browser Objective 1.1.2	Veb-brauzer bilan tanishish Maqsad 1.1.2
Web browsers (or simply, browsers) are software applications that enable users to easily access, view and navigate web pages on the Internet. Browsers allow you to interact with websites and to experience the amazing rich media that is available on the World Wide Web.	Veb-brauzerlar (yoki shunchaki brauzerlar) foydalanuvchilarga Internetdagi veb-sahifalarga oson kirish, ularni ko‘rish va ular bo‘ylab navigatsiya qilish imkonini beradigan dasturiy ilovalardir. Brauzerlar veb-saytlar bilan o‘zaro ishlash va World Wide Web’dagi boy multimedia kontentidan foydalanish imkonini beradi.
You may be familiar with several browsers, such as Microsoft Edge, Mozilla Firefox, Google Chrome, Apple Safari and Opera.	Siz Microsoft Edge, Mozilla Firefox, Google Chrome, Apple Safari va Opera kabi bir nechta brauzerlar bilan tanish bo‘lishingiz mumkin.
While a browser’s primary function is to retrieve pages from a web server and display them on your screen, you should understand that browsers are complex software applications that provide a number of services. They can be configured to suit your working style and keep you safe while you are online.	Brauzerning asosiy vazifasi veb-serverdan sahifalarni olib, ularni ekranda ko‘rsatish bo‘lsa-da, brauzerlar ko‘plab xizmatlarni taqdim etuvchi murakkab dasturiy ilovalar ekanini tushunish kerak. Ularni ish uslubingizga moslab sozlash va Internetda ishlayotganingizda xavfsizligingizni ta’minlash mumkin.
Using the Address Bar	Manzil satridan foydalanish
To visit a web page, you enter its URL into the browser address bar and press ENTER. The browser requests the page from the web server, and when it is received, the browser displays the page within the browser window. The address bar displays the URL of the page currently displayed in the browser window. If the webpage includes a title, it is displayed in the window tab.	Veb-sahifaga kirish uchun uning URL manzilini brauzerning manzil satriga kiritib, ENTER tugmasini bosib. Brauzer sahifani veb-serverdan so‘raydi va sahifa kelgach, uni brauzer oynasida ko‘rsatadi. Manzil satrida ayni paytda brauzer oynasida ko‘rsatilayotgan sahifaning URL manzili aks etadi. Agar veb-sahifaning sarlavhasi bo‘lsa, u oyna ichidagi tabda ko‘rsatiladi.
Window tab • Address bar • Browser window title bar	Oyna tabi • Manzil satri • Brauzer oynasining sarlavha satri
You can visit any web page by typing its URL into the address bar and pressing ENTER.	Manzil satriga URL manzilni kiritib, ENTER tugmasini bosish orqali istalgan veb-sahifaga kirishingiz mumkin.
As you navigate to other pages within a website (or as you visit pages on other websites), the URL shown in the address field updates to show the address of the current page. Most browsers maintain a history of URLs that you can access from the address bar. Clicking a URL displayed in the address bar history list has the same effect as entering the URL directly into the address bar.	Veb-sayt ichidagi boshqa sahifalarga o‘tganingizda yoki boshqa saytlarga tashrif buyurganingizda, manzil maydonidagi URL joriy sahifa manzilini ko‘rsatish uchun yangilanadi. Ko‘pchilik brauzerlar manzil satridan foydalanib kirish mumkin bo‘lgan URL tarixini saqlaydi. Manzil satridagi tarix ro‘yxatida ko‘rsatilgan URL’ni bosish URL’ni manzil satriga qo‘lda kiritish bilan bir xil natijani beradi.
In addition to displaying the URL, the address bar often includes several buttons. The Microsoft Edge address bar is shown in the previous screen.	URL’ni ko‘rsatishdan tashqari, manzil satrida odatda bir nechta tugma ham bo‘ladi. Microsoft Edge manzil satri oldingi ekranda ko‘rsatilgan.
Browser Scroll Bars	Brauzer aylantirish satrlari
Browser windows support horizontal and vertical scroll bars that appear when a window is too small to display all the contents of a web page. The following figure shows a browser window with a vertical scroll bar.	Brauzer oynalari gorizontal va vertikal aylantirish satrlarini qo‘llab-quvvatlaydi. Ular oyna veb-sahifaning barcha mazmunini ko‘rsatish uchun juda kichik bo‘lganda paydo bo‘ladi. Keyingi rasmda vertikal aylantirish satriga ega brauzer oynasi ko‘rsatilgan.

Study Note / O‘quv eslatmasi

- Software — Dasturiy ta’minot
- Internet — Internet
- Browser — Brauzer

Quick Check / Tezkor nazorat

10. What is the main concept explained in this section? / Ushbu bo‘limning asosiy tushunchasi nima?
11. Which English technical terms should be remembered? / Qaysi texnik English terminlarini yodlash kerak?
12. Can you explain the concept in Uzbek without looking at the text? / Matnga qaramasdan tushunchani o‘zbekcha izohlay olasizmi?

LESSON 1 — TECHNOLOGY BASICS | SECTION 5

Manba sahifasi / Source page: 5

IC3 Global Standard 6 – Level 1

Lesson 1: Technology Basics

Identifying Devices

Users interact with a computer through input and output devices.

- **Input** devices allow you to send information to the computer
- **Output** devices display (or otherwise send) information from the computer to you.

Keyboards and pointing devices are basic input devices. Monitors, printers, and speakers are basic output devices. Touch screens serve as both input and output devices because you can touch the screen to input information, and the screen displays the output.

Input Devices

Objective 1.2.1

Keyboards

The keyboard is the primary tool for sending information to the computer. You use it to enter data or to run commands in an application. Keyboards can be physical or virtual. In computing, a *virtual* device is one that does not physically exist, but is made to appear and act as if it exists by software. Virtual keyboards (and dial pads) are presented on screen in touch screen devices; the user “types” or dials by tapping the virtual keys that appear on the screen.

Keyboards come in many sizes and configurations. Some are ergonomically designed to protect against repetitive strain injuries such as carpal tunnel syndrome. Many include buttons for enhancing your multimedia experience.



The preceding figure shows a standard Windows keyboard, which includes the following types of keys:

- **Typewriter keys** – use these keys to type text and enter commands.

English — O‘zbekcha

ENGLISH	O‘ZBEK
ENGLISH	O‘ZBEK
Identifying Devices	Qurilmalarni aniqlash
Users interact with a computer through input and output devices. • Input devices allow you to send information to the computer. • Output devices display (or otherwise send) information from the computer to you.	Foydalanuvchilar kompyuter bilan kiritish va chiqarish qurilmalari orqali o‘zaro ishlaydi. • Kiritish qurilmalari ma’lumotni kompyuterga yuborish imkonini beradi. • Chiqarish qurilmalari kompyuterdagi ma’lumotni foydalanuvchiga ko‘rsatadi yoki boshqa usulda yetkazadi.
Keyboards and pointing devices are basic input devices. Monitors, printers, and speakers are basic output devices. Touch screens serve as both input and output devices because you can touch the screen to input information, and the screen displays the output.	Klaviatura va ko‘rsatish qurilmalari asosiy kiritish qurilmalaridir. Monitorlar, printerlar va karnaylar asosiy chiqarish qurilmalaridir. Sensorli ekranlar ham kiritish, ham chiqarish qurilmasi vazifasini bajaradi, chunki ekranga tegib ma’lumot kiritish mumkin va ekran natijani ko‘rsatadi.
Input Devices Objective 1.2.1 Keyboards	Kiritish qurilmalari Maqsad 1.2.1 Klaviaturalar
The keyboard is the primary tool for sending information to the computer. You use it to enter data or to run commands in an application. Keyboards can be physical or virtual. In computing, a virtual device is one that does not physically exist, but is made to appear and act as if it exists by software.	Klaviatura kompyuterga ma’lumot yuborishning asosiy vositasidir. Undan ma’lumot kiritish yoki ilovadagi buyruqlarni ishga tushirish uchun foydalanasiz. Klaviaturalar jismoniy yoki virtual bo‘lishi mumkin. Kompyuter texnologiyalarida virtual qurilma jismonan mavjud bo‘lmagan, ammo dasturiy ta’minot orqali mavjuddek ko‘rinadigan va ishlaydigan qurilmadir.
Virtual keyboards (and dial pads) are presented on screen in touch screen devices; the user “types” or dials by tapping the virtual keys that appear on the screen.	Virtual klaviaturalar (va raqam terish panellari) sensorli ekranli qurilmalarda ekranda ko‘rsatiladi; foydalanuvchi ekranda paydo bo‘ladigan virtual tugmalarga tegish orqali “yozadi” yoki raqam teradi.
Keyboards come in many sizes and configurations. Some are ergonomically designed to protect against repetitive strain injuries such as carpal tunnel syndrome. Many include buttons for enhancing your multimedia experience.	Klaviaturalar turli o‘lcham va konfiguratsiyalarda bo‘ladi. Ayrimlari karpal tunnel sindromi kabi takroriy zo‘riqishdan kelib chiqadigan jarohatlarning oldini olish uchun ergonomik tarzda ishlab chiqilgan. Ko‘pchiligida multimedia bilan ishlashni qulaylashtiruvchi tugmalar mavjud.
Function Keys • Escape • Numeric Keypad • Control • Alt • Cursor Movement • Windows	Funksiya tugmalari • Escape • Raqamli klaviatura • Control • Alt • Kursorni boshqarish • Windows
The preceding figure shows a standard Windows keyboard, which includes the following types of keys: • Typewriter keys – use these keys to type text and enter commands.	Yuqoridagi rasmda quyidagi turdagi tugmalarga ega standart Windows klaviaturasi ko‘rsatilgan: • Yozuv mashinkasi tugmalari — matn yozish va buyruqlar kiritish uchun ishlatiladi.

Study Note / O‘quv eslatmasi

- Software — Dasturiy ta’minot
- Input device — Kiritish qurilmasi
- Output device — Chiqarish qurilmasi

Quick Check / Tezkor nazorat

13. What is the main concept explained in this section? / Ushbu bo‘limning asosiy tushunchasi nima?
14. Which English technical terms should be remembered? / Qaysi texnik English terminlarini yodlash kerak?
15. Can you explain the concept in Uzbek without looking at the text? / Matnga qaramasdan tushunchani o‘zbekcha izohlay olasizmi?

LESSON 1 — TECHNOLOGY BASICS | SECTION 6

Manba sahifasi / Source page: 6

IC3 Global Standard 6 – Level 1

Lesson 1: Technology Basics

- Version 3.1 Generation 1 is almost identical to USB 3.0, except that it supports new connectors. The usual color coding for USB 3.0 ports and connectors is blue. Version 3.1 Generation 2 promises speeds of up to 10Gbps. The usual color coding for USB 3.1 ports and connectors is teal blue.

Most desktop systems sold today include several USB 2.0 and 3.0 ports.

Laptops generally include only 2 or 3 USB ports. If you want to connect several USB devices to a laptop, you can increase the number of available ports by connecting a USB hub. The hub uses one USB port on the computer, and provides, four, six, or eight open ports for connecting devices.

As portable devices have become smaller, the USB standard has adopted different sized connectors.

- USB 2.0 and 3.0 connectors are the flat rectangular connectors you are probably most familiar with.
- Mini-USB and Micro-USB connectors are commonly found on devices with a slim profile, with USB-C being the preferred connector on more recent devices.
- Lightning connectors were introduced for use with Apple products such as the iPhone, iPod, or iPad. If your device ships with a USB cable, it will include the correct connector type.



Bluetooth



Bluetooth is a wireless technology used to allow devices (such as computers and phones) to work with Bluetooth-enabled accessories (such as headsets, mice, keyboards, and so on).

Bluetooth devices and accessories must be "paired" with each other before they can communicate.

Bluetooth is also commonly used to enable hands-free phone calling by using a car's stereo system to make or receive calls without touching the phone, and to transfer files between devices that are in close proximity to each other.

Compatibility Considerations

It is important to understand that devices and accessories must be compatible. In other words, just because a device and an accessory are Bluetooth-enabled, it does not necessarily mean that they will work together.

For example, prior to iOS 13.4, iPads and iPhones supported an array of Bluetooth accessories such as headsets, remote controls, and keyboards; however, they did not support Bluetooth mice.

Case Study 3

The physical parts of a computer are called *hardware*. Hardware can be found connected to or inside the computer. Hardware that is connected to a computer is called a *peripheral device*. Devices that send information to the computer are *input* devices; those that receive output from the computer are *output* devices. Which of the following peripheral devices are input devices? Select the three options that apply.

- Scanner
- Monitor
- Keyboard
- Microphone
- Headphones
- Projector
- Speakers

English — O‘zbekcha

ENGLISH	O‘ZBEK
ENGLISH	O‘ZBEK
Version 3.1 Generation 1 is almost identical to USB 3.0, except that it supports new connectors. The usual color coding for USB 3.0 ports and connectors is blue. Version 3.1 Generation 2 promises speeds of up to 10Gbps. The usual color coding for USB 3.1 ports and connectors is teal blue.	3.1-versiya, 1-avlod USB 3.0 ga deyarli bir xil, faqat yangi ulagichlarni qo‘llab-quvvatlaydi. USB 3.0 portlari va ulagichlari uchun odatiy rang belgisi — ko‘k. 3.1-versiya, 2-avlod 10 Gbit/s gacha tezlikni va‘da qiladi. USB 3.1 portlari va ulagichlari uchun odatiy rang belgisi — ko‘kimtir-yashil (teal blue).
Most desktop systems sold today include several USB 2.0 and 3.0 ports.	Bugungi kunda sotiladigan ko‘pchilik stol kompyuterlarida bir nechta USB 2.0 va USB 3.0 portlari mavjud.
Laptops generally include only 2 or 3 USB ports. If you want to connect several USB devices to a laptop, you can increase the number of available ports by connecting a USB hub. The hub uses one USB port on the computer, and provides four, six, or eight open ports for connecting devices.	Noutbuklarda odatda faqat 2 yoki 3 ta USB port bo‘ladi. Agar noutbukga bir nechta USB qurilmasini ulashni istasangiz, USB hub ulash orqali mavjud portlar sonini oshirishingiz mumkin. Hub kompyuterdagi bitta USB portdan foydalanadi va qurilmalarni ulash uchun 4, 6 yoki 8 ta qo‘shimcha ochiq port taqdim etadi.
As portable devices have become smaller, the USB standard has adopted different sized connectors. • USB 2.0 and 3.0 connectors are the flat rectangular connectors you are probably most familiar with. • Mini-USB and Micro-USB connectors are commonly found on devices with a slim profile, with USB-C being the preferred connector on more recent devices. • Lightning connectors were introduced for use with Apple products such as the iPhone, iPod, or iPad. If your device ships with a USB cable, it will include the correct connector type.	Ko‘chma qurilmalar kichrayib borgan sari USB standarti turli o‘lchamdagi ulagichlarni qabul qildi. • USB 2.0 va 3.0 ulagichlari ko‘pchilikka tanish bo‘lgan yassi, to‘g‘ri to‘rtburchak shakldagi ulagichlardir. • Mini-USB va Micro-USB ulagichlari yupqa korpusli qurilmalarda keng uchraydi; zamonaviyroq qurilmalarda esa USB-C afzal ulagich hisoblanadi. • Lightning ulagichlari iPhone, iPod yoki iPad kabi Apple mahsulotlarida foydalanish uchun joriy etilgan. Agar qurilmangiz USB kabeli bilan yetkazib berilsa, unda mos ulagich turi bo‘ladi.
Bluetooth	Bluetooth
Bluetooth is a wireless technology used to allow devices (such as computers and phones) to work with Bluetooth-enabled accessories (such as headsets, mice, keyboards, and so on).	Bluetooth — kompyuter va telefonlar kabi qurilmalarni garnitura, sichqoncha, klaviatura va boshqa Bluetooth-ni qo‘llab-quvvatlovchi aksessuarlar bilan ishlashga imkon beruvchi simsiz texnologiya.
Bluetooth devices and accessories must be “paired” with each other before they can communicate.	Bluetooth qurilmalari va aksessuarlari aloqa qila olishidan oldin bir-biri bilan “juftlashtirilishi” (paired) kerak.
Bluetooth is also commonly used to enable hands-free phone calling by using a car’s stereo system to make or receive calls without touching the phone, and to transfer files between devices that are in close proximity to each other.	Bluetooth, shuningdek, telefonni qo‘lga olmasdan avtomobilning stereo tizimi orqali qo‘ng‘iroq qilish yoki qabul qilish, shuningdek bir-biriga yaqin joylashgan qurilmalar o‘rtasida fayl almashish uchun ham keng qo‘llanadi.
Compatibility Considerations	Moslik bo‘yicha muhim jihlatlar
It is important to understand that devices and accessories must be compatible. In other words, just because a device and an accessory are Bluetooth-enabled, it does not necessarily mean that they will work together.	Qurilmalar va aksessuarlar o‘zaro mos bo‘lishi kerakligini tushunish muhim. Boshqacha aytganda, qurilma va aksessuar Bluetooth-ni qo‘llab-quvvatlashi ularning albatta birgalikda ishlashini anglatmaydi.
For example, prior to iOS 13.4, iPads and iPhones supported an array of Bluetooth accessories such as headsets, remote controls, and keyboards; however, they did not support Bluetooth mice.	Masalan, iOS 13.4 dan oldin iPad va iPhone qurilmalari garnitura, masofadan boshqarish pultlari va klaviaturalar kabi turli Bluetooth aksessuarlarini qo‘llab-quvvatlagan, biroq Bluetooth sichqonchalarini qo‘llab-quvvatlamagan.
Case Study 3 The physical parts of a computer are called hardware. Hardware can be found connected to or inside the computer. Hardware that is connected to a computer is called a peripheral device. Devices that send information to the computer are input devices; those that receive output from the computer are output devices. Which of the following peripheral devices are input devices? Select the three options that apply. a. Scanner b. Monitor c. Keyboard d. Microphone e. Headphones f. Projector	3-amaliy holat Kompyuterning jismoniy qismlari apparat ta‘minoti deb ataladi. Apparat ta‘minoti kompyuterga ulangan holda yoki uning ichida bo‘lishi mumkin. Kompyuterga ulangan apparat qurilmasi periferik qurilma deb ataladi. Kompyuterga ma‘lumot yuboradigan qurilmalar kiritish qurilmalari; kompyuterdan chiqayotgan ma‘lumotni qabul qiladigan qurilmalar esa chiqarish qurilmalardir. Quyidagi periferik qurilmalardan qaysilari kiritish qurilmalari? Mos keladigan uchta variantni tanlang. a. Skaner b. Monitor c. Klaviatura d. Mikrofon e. Quloqchinlar

g. Speakers	f. Proyektor g. Karnaylar
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Study Note / O'quv eslatmasi

- Hardware — Apparat ta'minoti / qurilmalarning jismoniy qismi
- Input device — Kiritish qurilmasi
- Output device — Chiqarish qurilmasi
- Peripheral device — Tashqi/periferik qurilma
- Bluetooth — Bluetooth simsiz aloqa texnologiyasi

Quick Check / Tezkor nazorat

16. What is the main concept explained in this section? / Ushbu bo'limning asosiy tushunchasi nima?
17. Which English technical terms should be remembered? / Qaysi texnik English terminlarini yodlash kerak?
18. Can you explain the concept in Uzbek without looking at the text? / Matnga qaramasdan tushunchani o'zbekcha izohlay olasizmi?

LESSON 1 — TECHNOLOGY BASICS | SECTION 7

Manba sahifasi / Source page: 7

Lesson 1: Technology Basics

IC3 Global Standard 6 – Level 1

Software

Objectives 1.3.1, 1.3.3

While hardware is what makes our computers powerful and capable machines, it is software (programs) that makes our computing devices useful. The operating system provides an interface with the hardware itself, and software programs are what we use to get work done.

Software programs enable us to produce documents and charts; solve complex equations; calculate rocket trajectory; make online purchases; edit digital movies; decode the human genome; and many other menial or extraordinary things. Additionally, light-weight apps can keep us on schedule, remind us of appointments, or help us find our way to a location when construction crews block our familiar path. Software and apps keep us productive and empowered.

Locally-installed software is installed on a device and runs directly on that device. These programs are designed to work with specific operating systems and with hardware that meets certain minimum requirements.

Cloud-based software on the other hand runs on a dedicated server (not on your computer) and is designed to be accessed over the Internet. You will learn about cloud-based software in upcoming lessons.

Application Programs

An application program must be compatible with your hardware and operating system, and programs have specific requirements for processor power, video resolution, and memory. But how do you decide which application programs you should use?

Application programs vary in price from free to very expensive for those with large budgets. Deciding where to invest your money is not always an easy task, and understanding the different types of application programs available can help you get a clear grasp on what is available to fit your needs and your budget.

A productivity program is one that is dedicated to producing information, such as documents, presentations, worksheets, databases, charts, graphs, digital images, websites, electronic music or digital video. Different types of programs are suited for different types of tasks.

It is important to select the appropriate software program for the task at hand. Many programs share a number of features; to choose the one that best suits your needs, you must look closely at what you want to accomplish. It is recommended to make a list of all the tasks you need to perform, and check off items as you compare.

English — O'zbekcha

ENGLISH	O'ZBEK
ENGLISH	O'ZBEK
Software Objectives 1.3.1, 1.3.3	Dasturiy ta'minot Maqsadlar 1.3.1, 1.3.3
While hardware is what makes our computers powerful and capable machines, it is software (programs) that makes our computing devices useful. The operating system provides an interface with the hardware itself, and software programs are what we use to get work done.	Apparat ta'minoti kompyuterlarimizni kuchli va imkoniyatlarga boy mashinalarga aylantirsa, aynan dasturiy ta'minot (dasturlar) hisoblash qurilmalarini foydali qiladi. Operatsion tizim apparat ta'minoti bilan interfeysni ta'minlaydi, dasturiy ta'minot dasturlari esa ishlarni bajarish uchun foydalanadigan vositalarimizdir.
Software programs enable us to produce documents and charts; solve complex equations; calculate rocket trajectory; make online purchases; edit digital movies; decode the human genome; and many other menial or extraordinary things. Additionally, light-weight apps can keep us on schedule, remind us of appointments, or help us find our way to a location when construction crews block our familiar path. Software and apps keep us productive and empowered.	Dasturiy ta'minot hujjatlar va diagrammalar yaratish, murakkab tenglamalarni yechish, raketa trayektoriyasini hisoblash, onlayn xarid qilish, raqamli filmlarni tahrirlash, inson genomini dekodlash va boshqa oddiy yoki murakkab ishlarni bajarish imkonini beradi. Bundan tashqari, yengil ilovalar jadvalimizni tartibda saqlashi, uchrashuvlarni eslatishi yoki qurilish ishlari odatiy yo'limizni to'sib qo'yganda manzilni topishga yordam berishi mumkin. Dastur va ilovalar bizni samarali va imkoniyatli qiladi.
Locally-installed software is installed on a device and runs directly on that device. These programs are designed to work with specific operating systems and with hardware that meets certain minimum requirements.	Mahalliy o'rnatiladigan dasturiy ta'minot qurilmaga o'rnatiladi va bevosita shu qurilmada ishlaydi. Bunday dasturlar muayyan operatsion tizimlar va ma'lum minimal talablarga javob beradigan apparat bilan ishlash uchun yaratiladi.
Cloud-based software on the other hand runs on a dedicated server (not on your computer) and is designed to be accessed over the Internet. You will learn about cloud-based software in upcoming lessons.	Bulutga asoslangan dasturiy ta'minot esa maxsus serverda (kompyuteringizning o'zida emas) ishlaydi va Internet orqali foydalanish uchun mo'ljallangan. Bulutga asoslangan dasturiy ta'minot haqida keyingi darslarda batafsil o'rganasiz.
Application Programs	Amaliy dasturlar
An application program must be compatible with your hardware and operating system, and programs have specific requirements for processor power, video resolution, and memory. But how do you decide which application programs you should use?	Amaliy dastur apparat ta'minotingiz va operatsion tizimingiz bilan mos bo'lishi kerak. Dasturlarning protsessor quvvati, video aniqligi va xotiraga nisbatan muayyan talablari mavjud. Ammo qaysi amaliy dasturlardan foydalanishni qanday tanlash mumkin?
Application programs vary in price from free to very expensive for those with large budgets. Deciding where to invest your money is not always an easy task, and understanding the different types of application programs available can help you get a clear grasp on what is available to fit your needs and your budget.	Amaliy dasturlar narxi bepul variantlardan katta budjet talab qiladigan juda qimmat dasturlargacha farq qiladi. Mablag'ni qayerga sarflashni tanlash har doim ham oson emas. Mavjud amaliy dasturlar turlarini tushunish ehtiyojlaringiz va budjetingizga mos variantlarni aniqroq baholashga yordam beradi.
A productivity program is one that is dedicated to producing information, such as documents, presentations, worksheets, databases, charts, graphs, digital images, websites, electronic music or digital video. Different types of programs are suited for different types of tasks.	Mahsuldorlik dasturi — hujjatlar, taqdimotlar, ish varaqlari, ma'lumotlar bazalari, diagrammalar, grafiklar, raqamli tasvirlar, veb-saytlar, elektron musiqa yoki raqamli video kabi axborotlarni yaratishga mo'ljallangan dasturdir. Turli vazifalar uchun dasturlarning turli turlari mos keladi.
It is important to select the appropriate software program for the task at hand. Many programs share a number of features; to choose the one that best suits your needs, you must look closely at what you want to accomplish. It is recommended to make a list of all the tasks you need to perform, and check off items as you compare.	Bajariladigan vazifaga mos dasturiy ta'minotni tanlash muhim. Ko'plab dasturlar bir qator umumiy funksiyalarga ega; ehtiyojingizga eng mosini tanlash uchun nimaga erishmoqchi ekaningizni diqqat bilan ko'rib chiqishingiz kerak. Bajarishingiz kerak bo'lgan barcha vazifalar ro'yxatini tuzib, dasturlarni taqqoslash davomida mos bandlarni belgilab borish tavsiya etiladi.

Study Note / O'quv eslatmasi

- Hardware — Apparat ta'minoti / qurilmalarning jismoniy qismi
- Software — Dasturiy ta'minot
- Application program — Amaliy dastur
- Internet — Internet
- Cloud-based software — Bulutga asoslangan dasturiy ta'minot

Quick Check / Tezkor nazorat

19. What is the main concept explained in this section? / Ushbu bo‘limning asosiy tushunchasi nima?
20. Which English technical terms should be remembered? / Qaysi texnik English terminlarini yodlash kerak?
21. Can you explain the concept in Uzbek without looking at the text? / Matnga qaramasdan tushunchani o‘zbekcha izohlay olasizmi?

LESSON 1 — TECHNOLOGY BASICS | SECTION 8

Manba sahifasi / Source page: 8

IC3 Global Standard 6 – Level 1

Lesson 1: Technology Basics

Word Processing

A *word processing* application enables you to create, edit, save, format and print documents. Most word processing programs also allow you to insert pictures and other digital objects. You can use a word processing application to create letters, memos, invoices, faxes, articles, and sometimes basic Web pages, newsletters, forms, brochures, or flyers.

The following examples of documents you can produce with word processing programs include a letter, a newsletter, and a flyer.



Popular word processing applications include:

Microsoft Word – by Microsoft. Word is a popular word processing application that you can use to type and format text and images for a variety of documents. Advanced features help you create tables, indices, and outlines. It is often installed on Apple devices as well, and is available in a web version.

WordPerfect – by Corel. WordPerfect was one of the first highly successful word processors for DOS and Windows based systems, and it has also been available for a wide variety of operating systems.

Pages for Mac – by Apple. Pages for Mac is a powerful word processor that you can use to create documents on Mac and on iOS devices. Design tools allow you to create charts, tables, bi-directional text and more.

Spreadsheets

Spreadsheet programs perform functions and mathematical calculations, help you analyze data, and produce graphs, charts, and diagrams. If you need to track numbers or analyze information for trends or patterns, a spreadsheet is a good choice.

Spreadsheets provide the ability to sort, find or filter information, and are very useful for managing and manipulating large amounts of data. You can use spreadsheets to analyze and present information such as expenses, assignment or report marks, financial forecasts, cash flow and so on, as shown in the above image.

Today's date	27-Apr-18	Month	Payment in £	Exchange	Payment in \$	Month	Exchange Rate
Amount to invest	\$200,000	1	\$28,666.67	1.57	\$43,966.67	0	1.59
Amount to invest	\$334,966.10	2	\$28,666.67	1.5	\$43,000.00	1	1.57
Annual interest rate	3.00%	3	\$28,666.67	1.52	\$43,533.33	2	1.50
Monthly payment							

Country	Awards by Country	Longest song	Year and name of the song
United States of America	89		7:02 1989 We Are the World
England	7		
Ireland	2	Shortest song	Last name of singer / band
Canada	1		3:09 Harris
Mexico	1		
New Zealand	1	Average length of Winning Songs	
Totals	100		

COUNTRIES WITH THE MOST GRAMMY AWARDS	
NEW ZEALAND	89
MEXICO	89
CANADA	89
HOLLAND	89
ENGLAND	89

Popular spreadsheet programs include:

Microsoft Excel – by Microsoft. Excel is part of the Microsoft 365 suite. It is a spreadsheet application that runs on Windows, Mac OS X, Android, and iOS. It features calculation, graphing tools, pivot tables, and a macro programming language called Visual Basic for Applications.

Google Sheets – by Google. Sheets is a web-based spreadsheet program that is a part of the larger Google Workspace (formerly known as G Suite or Google Docs Editor Suite). Being web-based, Google Sheets facilitates collaborative work by 'inviting' or sharing with others to view and/or edit the spreadsheets in real-time (more on collaborative work programs in later lessons).

English — O‘zbekcha

ENGLISH	O‘ZBEK
ENGLISH	O‘ZBEK
Word Processing	Matnni qayta ishlash
A word processing application enables you to create, edit, save, format and print documents. Most word processing programs also allow you to insert pictures and other digital objects. You can use a word processing application to create letters, memos, invoices, faxes, articles, and sometimes basic Web pages, newsletters, forms, brochures, or flyers.	Matn protsessori ilovasi hujjatlarni yaratish, tahrirlash, saqlash, formatlash va chop etish imkonini beradi. Ko‘pchilik matn protsessoralari rasmlar va boshqa raqamli obyektlarni ham qo‘shishga imkon beradi. Bunday dastur yordamida xatlar, xizmat yozuvlari, hisob-fakturalar, fakslar, maqolalar, ba‘zan oddiy veb-sahifalar, axborot byulletenlari, shakllar, bukletlar yoki flayerlar yaratish mumkin.
The following examples of documents you can produce with word processing programs include a letter, a newsletter, and a flyer.	Matn protsessori yordamida yaratish mumkin bo‘lgan hujjatlarga xat, axborot byulleteni va flayer misol bo‘la oladi.
Popular word processing applications include: Microsoft Word – by Microsoft. Word is a popular word processing application that you can use to type and format text and images for a variety of documents. Advanced features help you create tables, indices, and outlines. It is often installed on Apple devices as well, and is available in a web version. WordPerfect – by Corel. WordPerfect was one of the first highly successful word processors for DOS and Windows based systems, and it has also been available for a wide variety of operating systems. Pages for Mac – by Apple. Pages for Mac is a powerful word processor that you can use to create documents on Mac and on iOS devices. Design tools allow you to create charts, tables, bi-directional text and more.	Mashhur matn protsessori ilovalari: Microsoft Word — Microsoft kompaniyasining dasturi. Word turli hujjatlar uchun matn va rasmlarni kiritish hamda formatlashda ishlatiladigan mashhur matn protsessori. Kengaytirilgan funksiyalar jadvallar, indekslar va rejalar yaratishga yordam beradi. U ko‘pincha Apple qurilmalariga ham o‘rnatiladi va veb-versiyasi ham mavjud. WordPerfect — Corel kompaniyasining dasturi. WordPerfect DOS va Windows tizimlari uchun ilk muvaffaqiyatli matn protsessoralardan biri bo‘lgan va turli operatsion tizimlarda ham mavjud bo‘lgan. Pages for Mac — Apple kompaniyasining dasturi. Pages for Mac — Mac va iOS qurilmalarida hujjatlar yaratish mumkin bo‘lgan kuchli matn protsessori. Dizayn vositalari diagrammalar, jadvallar, ikki yo‘nalishli matn va boshqa obyektlarni yaratish imkonini beradi.
Spreadsheets	Elektron jadvallar
Spreadsheet programs perform functions and mathematical calculations, help you analyze data, and produce graphs, charts, and diagrams. If you need to track numbers or analyze information for trends or patterns, a spreadsheet is a good choice.	Elektron jadval dasturlari funksiyalar va matematik hisob-kitoblarni bajaradi, ma‘lumotlarni tahlil qilishga yordam beradi hamda grafiklar, diagrammalar va chizmalar yaratadi. Agar raqamlarni kuzatish yoki axborotni tendensiyalar va qonuniyatlar bo‘yicha tahlil qilish kerak bo‘lsa, elektron jadval yaxshi tanlovdir.
Spreadsheets provide the ability to sort, find or filter information, and are very useful for managing and manipulating large amounts of data. You can use spreadsheets to analyze and present information such as expenses, assignment or report marks, financial forecasts, cash flow and so on.	Elektron jadvallar ma‘lumotlarni saralash, topish yoki filtrlash imkonini beradi va katta hajmdagi ma‘lumotlarni boshqarish hamda qayta ishlashda juda foydalidir. Ular yordamida xarajatlar, topshiriq yoki hisobot baholari, moliyaviy prognozlar, pul oqimi va boshqa ma‘lumotlarni tahlil qilish va taqdim etish mumkin.
Popular spreadsheet programs include: Microsoft Excel – by Microsoft. Excel is part of the Microsoft 365 suite. It is a spreadsheet application that runs on Windows, Mac OS X, Android, and iOS. It features calculation, graphing tools, pivot tables, and a macro programming language called Visual Basic for Applications. Google Sheets – by Google. Sheets is a web-based spreadsheet program that is a part of the larger Google Workspace (formerly known as G Suite or Google Docs Editor Suite). Being web-based, Google Sheets facilitates collaborative work by ‘inviting’ or sharing with others to view and/or edit the spreadsheets in real-time.	Mashhur elektron jadval dasturlari: Microsoft Excel — Microsoft kompaniyasining dasturi. Excel Microsoft 365 to‘plamining bir qismidir. U Windows, Mac OS X, Android va iOS’da ishlaydigan elektron jadval ilovasidir. Unda hisoblash vositalari, grafik yaratish vositalari, pivot jadvallar va Visual Basic for Applications deb ataladigan makrodasturlash tili mavjud. Google Sheets — Google kompaniyasining dasturi. Sheets Google Workspace tarkibiga kiruvchi veb-asosli elektron jadval dasturidir (ilgari G Suite yoki Google Docs Editor Suite deb atalgan). Veb-asosli bo‘lgani sababli Google Sheets boshqalarni real vaqt rejimida jadvallarni ko‘rish yoki tahrirlashga taklif qilish yoki ulashish orqali hamkorlikda ishlashni osonlashtiradi.

Study Note / O‘quv eslatmasi

Ushbu bo‘limdagi asosiy atamalarni English va Uzbek ko‘rinishda birga yodlang.

Quick Check / Tezkor nazorat

22. What is the main concept explained in this section? / Ushbu bo‘limning asosiy tushunchasi nima?
23. Which English technical terms should be remembered? / Qaysi texnik English terminlarini yodlash kerak?
24. Can you explain the concept in Uzbek without looking at the text? / Matnga qaramasdan tushunchani o‘zbekcha izohlay olasizmi?

LESSON 1 — TECHNOLOGY BASICS | SECTION 9

Manba sahifasi / Source page: 9

Lesson 1: Technology Basics

IC3 Global Standard 6 – Level 1

EULA and Registration

Once the installation process begins, you will be asked to agree to an End User License Agreement (EULA); depending on the vendor, this may appear in a separate screen or it may be a link you should click in order to read the contents of the agreement. Typically, you agree not to make illegal copies of the software to distribute to others, and you agree not to hold the software vendor liable for any damage or expenses that may occur from misuse or improper use of the software, such as putting incorrect entries into an income tax preparation program which results in an audit by the tax collector.

When the installation is complete, you will usually be asked to register or activate your copy of the program. Performing this last step ensures that you will be notified of any updates to the program. It also usually provides the option to call technical support should you require technical assistance.

Case Study 4

Software applications are programs that tell the computer what to do, when to do it, and how to do it. Most software is available for download from the Internet.

Which statements below are TRUE about downloading? Select the two options that apply.

- Downloading is the same as streaming.
- Downloading requires an Internet connection.
- Downloading is when you receive files or data to your computer using the Internet.
- Downloading is when you send files or data from your computer using the Internet.

Recognizing Basic Hardware

Objective 1.4.1

Desktop Computers

Desktop computers are designed to sit on or beneath a desk. They are fairly large, although many are available in compact configurations so that they require less space.

Desktop computers come in two basic designs:

- **Macintosh or Macs** – produced by Apple. These machines feature a sleek design in which the monitor and processing unit are housed within the same case. Only Apple produces the Mac. Macs are designed to run the Mac OS X operating system.
- **Personal Computer or PC** – produced originally by IBM, and later produced by several manufacturers. Traditionally, in a desktop PC, the processing unit is housed within a case, and an external monitor, keyboard, and mouse are attached with cables. The fully assembled combination is sometimes referred to as a *work station*. Today, there are also some all-in-one units available. You can use a wide variety of operating systems on a PC, but by far the one most commonly installed on a PC is the Windows operating system.

Both Macs and PCs are available in a wide variety of types and configurations (including desktop models, laptops, servers, and touch screen devices).

English — O'zbekcha

ENGLISH	O'ZBEK
ENGLISH	O'ZBEK
EULA and Registration	EULA va ro'yxatdan o'tish
Once the installation process begins, you will be asked to agree to an End User License Agreement (EULA); depending on the vendor, this may appear in a separate screen or it may be a link you should click in order to read the contents of the agreement. Typically, you agree not to make illegal copies of the software to distribute to others, and you agree not to hold the software vendor liable for any damage or expenses that may occur from misuse or improper use of the software, such as putting incorrect entries into an income tax preparation program which results in an audit by the tax collector.	O'rnatish jarayoni boshlangach, sizdan Yakuniy foydalanuvchi litsenziya shartnomasiga (EULA) rozilik berish so'raladi. Dastur yetkazib beruvchiga qarab, bu alohida oynada ko'rsatilishi yoki shartnoma mazmunini o'qish uchun bosiladigan havola ko'rinishida bo'lishi mumkin. Odatda siz dasturdan noqonuniy nusxalar tayyorlab, ularni boshqalarga tarqatmaslikka hamda dasturdan noto'g'ri yoki maqsadsiz foydalanish natijasida yuzaga kelishi mumkin bo'lgan zarar va xarajatlar uchun dastur yetkazib beruvchini javobgar qilmaslikka rozilik bildirasiz; masalan, daromad solig'ini hisoblash dasturiga noto'g'ri ma'lumot kiritish soliq tekshiruviga olib kelishi mumkin.
When the installation is complete, you will usually be asked to register or activate your copy of the program. Performing this last step ensures that you will be notified of any updates to the program. It also usually provides the option to call technical support should you require technical assistance.	O'rnatish tugagach, odatda dastur nusxangizni ro'yxatdan o'tkazish yoki faollashtirish so'raladi. Bu oxirgi bosqichni bajarish dastur yangilanishlari haqida xabardor bo'lishingizni ta'minlaydi. Shuningdek, texnik yordam kerak bo'lsa, texnik ko'mak xizmatiga murojaat qilish imkonini beradi.
Case Study 4 Software applications are programs that tell the computer what to do, when to do it, and how to do it. Most software is available for download from the Internet. Which statements below are TRUE about downloading? Select the two options that apply. a. Downloading is the same as streaming. b. Downloading requires an Internet connection. c. Downloading is when you receive files or data to your computer using the Internet. d. Downloading is when you send files or data from your computer using the Internet.	4-amaliy holat Dasturiy ilovalar kompyuterga nima qilish, qachon qilish va qanday qilish kerakligini ko'rsatuvchi dasturlardir. Ko'pchilik dasturlarni Internet orqali yuklab olish mumkin. Quyidagi fikrlardan qaysilari yuklab olish (downloading) haqida TO'G'RI? Mos keladigan ikkita variantni tanlang. a. Yuklab olish oqim orqali uzatish (streaming) bilan bir xil. b. Yuklab olish uchun Internet ulanishi kerak. c. Yuklab olish — Internet orqali fayl yoki ma'lumotlarni kompyuteringizga qabul qilishdir. d. Yuklab olish — Internet orqali kompyuteringizdan fayl yoki ma'lumotlarni yuborishdir.
Recognizing Basic Hardware Objective 1.4.1	Asosiy apparat ta'minotini aniqlash Maqsad 1.4.1
Desktop Computers	Stol kompyuterlari
Desktop computers are designed to sit on or beneath a desk. They are fairly large, although many are available in compact configurations so that they require less space.	Stol kompyuterlari stol ustida yoki ostida joylashish uchun mo'ljallangan. Ular nisbatan katta bo'ladi, biroq kamroq joy egallashi uchun ixcham konfiguratsiyadagi ko'plab modellar ham mavjud.
Desktop computers come in two basic designs: • Macintosh or Macs – produced by Apple. These machines feature a sleek design in which the monitor and processing unit are housed within the same case. Only Apple produces the Mac. Macs are designed to run the Mac OS X operating system. • Personal Computer or PC – produced originally by IBM, and later produced by several manufacturers. Traditionally, in a desktop PC, the processing unit is housed within a case, and an external monitor, keyboard, and mouse are attached with cables. The fully assembled combination is sometimes referred to as a work station. Today, there are also some all-in-one units available. You can use a wide variety of operating systems on a PC, but by far the one most commonly installed on a PC is the Windows operating system.	Stol kompyuterlari ikki asosiy dizaynda bo'ladi: • Macintosh yoki Mac — Apple tomonidan ishlab chiqariladi. Bu qurilmalarda monitor va hisoblash bloki bitta korpus ichida joylashgan zamonaviy dizayn mavjud. Mac'ni faqat Apple ishlab chiqaradi. Mac kompyuterlari Mac OS X operatsion tizimida ishlash uchun mo'ljallangan. • Personal Computer yoki PC — dastlab IBM tomonidan ishlab chiqarilgan, keyinchalik ko'plab ishlab chiqaruvchilar tomonidan ishlab chiqarila boshlangan. An'anaviy stol kompyuterida hisoblash bloki korpus ichida joylashadi, tashqi monitor, klaviatura va sichqoncha esa kabellar orqali ulanadi. To'liq yig'ilgan bunday majmua ba'zan ish stansiyasi deb ataladi. Bugungi kunda all-in-one qurilmalar ham mavjud. PC'da turli operatsion tizimlardan foydalanish mumkin, ammo eng ko'p o'rnatiladigani Windows operatsion tizimidir.
Both Macs and PCs are available in a wide variety of types and configurations (including desktop models, laptops, servers, and touch screen devices).	Mac va PC'larning ikkalasi ham turli xil turlar va konfiguratsiyalarda, jumladan stol modellari, noutbuklar, serverlar va sensorli ekranli qurilmalar ko'rinishida mavjud.

Study Note / O'quv eslatmasi

- Hardware — Apparat ta'minoti / qurilmalarning jismoniy qismi
- Software — Dasturiy ta'minot
- Internet — Internet
- Download — Yuklab olish

Quick Check / Tezkor nazorat

25. What is the main concept explained in this section? / Ushbu bo'limning asosiy tushunchasi nima?
26. Which English technical terms should be remembered? / Qaysi texnik English terminlarini yodlash kerak?
27. Can you explain the concept in Uzbek without looking at the text? / Matnga qaramasdan tushunchani o'zbekcha izohlay olasizmi?

LESSON 1 — TECHNOLOGY BASICS | SECTION 10

Manba sahifasi / Source page: 10

IC3 Global Standard 6 – Level 1

Lesson 1: Technology Basics



PC



Mac

Desktop computers are stable and powerful and most include ample storage space (500GB – 1TB on average), a fair amount of installed memory (8GB – 16GB on average), the ability to read data stored on a wide variety of storage media, and the ability to play music and video. They include several USB ports for connecting external devices, and may include one or more slots for reading and writing to SD cards. Many are equipped with powerful video and sound cards. They usually include a network port for a wired network connection, and most include built-in wireless networking ability as well.

They are also easily upgradeable – you can open the case to access the internal components and there is generally enough “working room” to easily replace or upgrade components such as a video, audio, or network card. You can also easily add more storage space.

Laptop (or Notebook) Computers

Laptop (or notebook) computers are designed to be portable. They are small and light enough to sit on your lap. They are also self-contained with everything you need included in one unit – display, keyboard, camera, speakers, pointing device.

Laptops run the same operating systems as their desktop counterparts, and include the same internal components such as a hard drive and installed memory. Laptops also include a rechargeable battery that is charged from an AC adapter – these can range greatly in power.



Laptops are very popular with students and business people alike because of their portability. For example, students can bring a laptop to class and take notes, then take it home to do homework and other assignments.

English — O‘zbekcha

ENGLISH	O‘ZBEK
ENGLISH	O‘ZBEK
PC • Mac	PC • Mac
Desktop computers are stable and powerful and most include ample storage space (500GB – 1TB on average), a fair amount of installed memory (8GB - 16GB on average), the ability to read data stored on a wide variety of storage media, and the ability to play music and video. They include several USB ports for connecting external devices, and may include one or more slots for reading and writing to SD cards. Many are equipped with powerful video and sound cards. They usually include a network port for a wired network connection, and most include built-in wireless networking ability as well.	Stol kompyuterlari barqaror va kuchli bo‘lib, ko‘pchiligida yetarli saqlash joyi (o‘rtacha 500 GB–1 TB), ma‘lum hajmdagi o‘rnatilgan xotira (o‘rtacha 8–16 GB), turli saqlash vositalarida saqlangan ma‘lumotlarni o‘qish va musiqa hamda videoni ijro etish imkoniyati mavjud. Ularda tashqi qurilmalarni ulash uchun bir nechta USB portlar bo‘ladi, shuningdek SD-kartalarni o‘qish va yozish uchun bir yoki bir nechta slot bo‘lishi mumkin. Ko‘pchiligida kuchli video va ovoz kartalari mavjud. Odatda simli tarmoqqa ulanish uchun tarmoq porti bo‘ladi va ko‘pchiligida simsiz tarmoq imkoniyati ham ichki o‘rnatilgan.
They are also easily upgradeable – you can open the case to access the internal components and there is generally enough “working room” to easily replace or upgrade components such as a video, audio, or network card. You can also easily add more storage space.	Ularini yangilash ham oson: ichki komponentlarga kirish uchun korpusni ochish mumkin va odatda video, audio yoki tarmoq kartasi kabi komponentlarni almashtirish yoki yangilash uchun yetarli ish joyi bo‘ladi. Qo‘shimcha saqlash joyini ham oson qo‘shish mumkin.
Laptop (or Notebook) Computers	Noutbuk (yoki Notebook) kompyuterlari
Laptop (or notebook) computers are designed to be portable. They are small and light enough to sit on your lap. They are also self-contained with everything you need included in one unit - display, keyboard, camera, speakers, pointing device.	Noutbuklar ko‘chma bo‘lishi uchun ishlab chiqilgan. Ular tizzada ishlatish mumkin bo‘lgan darajada kichik va yengil. Ularning barchasi bitta qurilmada jamlangan: displey, klaviatura, kamera, karnaylar va ko‘rsatish qurilmasi.
Laptops run the same operating systems as their desktop counterparts, and include the same internal components such as a hard drive and installed memory. Laptops also include a rechargeable battery that is charged from an AC adapter – these can range greatly in power.	Noutbuklarda stol kompyuterlaridagi kabi operatsion tizimlar ishlaydi va qattiq disk hamda o‘rnatilgan xotira kabi ichki komponentlar mavjud. Noutbuklarda, shuningdek, AC adapter orqali zaryadlanadigan qayta zaryadlanuvchi batareya bo‘ladi; ularning quvvati modelga qarab sezilarli darajada farq qilishi mumkin.
Laptops are very popular with students and business people alike because of their portability. For example, students can bring a laptop to class and take notes, then take it home to do homework and other assignments.	Noutbuklar ko‘chma bo‘lgani sababli talabalar va biznes vakillari orasida juda mashhur. Masalan, talaba noutbukni darsga olib kelib konspekt yozishi, keyin esa uni uyiga olib borib uy vazifalari va boshqa topshiriqlarni bajarishi mumkin.

Study Note / O‘quv eslatmasi

- Network — Tarmoq

Quick Check / Tezkor nazorat

28. What is the main concept explained in this section? / Ushbu bo‘limning asosiy tushunchasi nima?
29. Which English technical terms should be remembered? / Qaysi texnik English terminlarini yodlash kerak?
30. Can you explain the concept in Uzbek without looking at the text? / Matnga qaramasdan tushunchani o‘zbekcha izohlay olasizmi?

LESSON 1 — TECHNOLOGY BASICS | SECTION 11

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Lesson 1: Technology Basics

IC3 Global Standard 6 – Level 1

Recently discontinued, but still available (no longer supported, or updated) mobile operating systems include:

- **Windows 10 Mobile** (formerly called Windows Phone) – is from Microsoft. It is closed-source and proprietary and it is integrated with Microsoft services such as OneDrive and Office, Xbox Music, Xbox Video, Xbox Live and Bing.
- **BlackBerry 10** – is from BlackBerry and is closed-source and proprietary. It runs only on phones and tablets manufactured by BlackBerry.

Popular Operating Systems

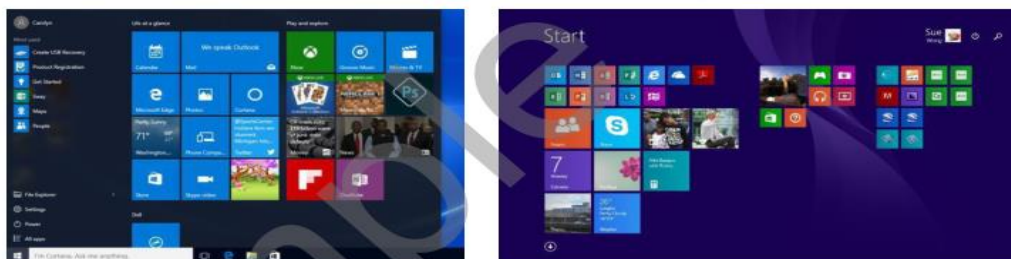
Objective 1.5.2

While there have been several versions of computer operating systems available, the most commonly used ones include the following:

Windows

Windows operating systems are designed by Microsoft. The current version is Windows 10 although there may be some home systems with Windows 8 installed.

The Windows 10 and Windows 8 Desktops are shown here:



Mac OS X

Mac OS X is designed by Apple for Macintosh computers. Recent versions are:

- Version 10.13 – High Sierra
- Version 10.14 – Mojave
- Version 10.15 – Catalina

The Mac OS X Catalina and Mojave desktops are shown here:

English — O‘zbekcha

ENGLISH	O‘ZBEK
ENGLISH	O‘ZBEK
Recently discontinued, but still available (no longer supported, or updated) mobile operating systems include:	Yaqinda qo‘llab-quvvatlashi yoki yangilanishi to‘xtatilgan, biroq hali ham mavjud bo‘lgan mobil operatsion tizimlarga quyidagilar kiradi:
• Windows 10 Mobile (formerly called Windows Phone) – is from Microsoft. It is closed-source and proprietary and it is integrated with Microsoft services such as OneDrive and Office, Xbox Music, Xbox Video, Xbox Live and Bing. • BlackBerry 10 – is from BlackBerry and is closed-source and proprietary. It runs only on phones and tablets manufactured by BlackBerry.	• Windows 10 Mobile (ilgari Windows Phone deb atalgan) — Microsoft kompaniyasiga tegishli. U yopiq kodli va xususiy dasturiy ta‘minot bo‘lib, OneDrive, Office, Xbox Music, Xbox Video, Xbox Live va Bing kabi Microsoft xizmatlari bilan integratsiyalashgan. • BlackBerry 10 — BlackBerry kompaniyasiga tegishli, yopiq kodli va xususiy tizimdir. U faqat BlackBerry tomonidan ishlab chiqarilgan telefon va planshetlarda ishlaydi.
Popular Operating Systems Objective 1.5.2	Mashhur operatsion tizimlar Maqsad 1.5.2
While there have been several versions of computer operating systems available, the most commonly used ones include the following:	Kompyuter operatsion tizimlarining ko‘plab versiyalari mavjud bo‘lgan bo‘lsa-da, eng ko‘p ishlatiladiganlariga quyidagilar kiradi:
Windows Windows operating systems are designed by Microsoft. The current version is Windows 10 although there may be some home systems with Windows 8 installed. The Windows 10 and Windows 8 Desktops are shown here:	Windows Windows operatsion tizimlari Microsoft tomonidan ishlab chiqiladi. Ushbu manbadagi ma‘lumotga ko‘ra, joriy versiya Windows 10 bo‘lib, ayrim uy kompyuterlarida Windows 8 o‘rnatilgan bo‘lishi mumkin. Bu yerda Windows 10 va Windows 8 Desktop ekranlari ko‘rsatilgan:
Mac OS X Mac OS X is designed by Apple for Macintosh computers. Recent versions are: • Version 10.13 – High Sierra • Version 10.14 – Mojave • Version 10.15 – Catalina The Mac OS X Catalina and Mojave desktops are shown here:	Mac OS X Mac OS X Apple tomonidan Macintosh kompyuterlari uchun ishlab chiqilgan. Yaqin davrdagi versiyalar: • 10.13-versiya — High Sierra • 10.14-versiya — Mojave • 10.15-versiya — Catalina Bu yerda Mac OS X Catalina va Mojave Desktop ekranlari ko‘rsatilgan:

Study Note / O‘quv eslatmasi

- Proprietary — Xususiy / yopiq litsenziyal

Quick Check / Tezkor nazorat

31. What is the main concept explained in this section? / Ushbu bo‘limning asosiy tushunchasi nima?
32. Which English technical terms should be remembered? / Qaysi texnik English terminlarini yodlash kerak?
33. Can you explain the concept in Uzbek without looking at the text? / Matnga qaramasdan tushunchani o‘zbekcha izohlay olasizmi?

LESSON 1 — TECHNOLOGY BASICS | SECTION 12

Manba sahifasi / Source page: 12

Lesson 1: Technology Basics

IC3 Global Standard 6 – Level 1

In computing, a network is simply an arrangement of computers (and additional computing devices) that are connected in such a way that they can communicate and share information. Individual networks can also be connected to other networks, and this practice is referred to as *internetworking*.

A network can consist of two or three computers in a single room, while an *internetwork* can consist of millions of computers connected across the globe.



Infrastructure Is Everything

Networking technologies have been in a state of constant development and improvement since the late 1960s. However, networking becomes practical and desirable and adopted into widespread use only when the proper infrastructure is in place.

Infrastructure is the basic underlying physical structure or framework needed for the operation of a service or an enterprise. In the case of networking, infrastructure is the hardware that supports high-speed communications and data transfer. Infrastructure is millions of miles of fiber-optic cable, high-powered cell relay towers dotting the landscape, thousands of communication satellites in Earth's orbit, high-speed routers and servers and exchange points strategically positioned along the Internet, and consumer devices capable of participating in modern networks.

It has taken over fifty years and billions of dollars to build the infrastructure that our modern networks rely on today. Modern networking technologies and networks evolve and expand at an exponential rate because finally there is now sufficient infrastructure to support that type of growth.

Connectedness is Key

You will learn about different types of networks in this course. Even though these networks are distinct from one another, they also inter-connect with each other. Cellular networks, the telephone network, and the Internet all *interconnect*. Since our private home, school, or businesses networks and devices tap into these large networks, we are all interconnected.

Why Network?

Networks are in wide use today because they are practical and useful. They enable users to share many things, including:

- **Files** – for example, Betty can create and store a file on her computer, and her fellow network user Barney can open and work with Betty's file without having to obtain a copy of the file.
- **Resources** – printers and scanners are commonly shared on a network. An entire office can share one or two printers, eliminating the need to purchase and attach a printer to each computer.
- **Internet connections** – networks within a home or office allow several users to share an Internet connection.

English — O‘zbekcha

ENGLISH	O‘ZBEK
ENGLISH	O‘ZBEK
In computing, a network is simply an arrangement of computers (and additional computing devices) that are connected in such a way that they can communicate and share information. Individual networks can also be connected to other networks, and this practice is referred to as internetworking. A network can consist of two or three computers in a single room, while an internetwork can consist of millions of computers connected across the globe.	Kompyuter texnologiyalarida tarmoq — kompyuterlar (va qo‘shimcha hisoblash qurilmalari) o‘zaro aloqa qilishi va axborot almashishi mumkin bo‘lgan tarzda ulangan tizimdir. Alohida tarmoqlar boshqa tarmoqlarga ham ulanishi mumkin va bu jarayon internetworking deb ataladi. Tarmoq bitta xonadagi ikki yoki uchta kompyuterdan iborat bo‘lishi mumkin, internetwork esa butun dunyo bo‘ylab ulangan millionlab kompyuterlarni o‘z ichiga olishi mumkin.
Infrastructure Is Everything	Infratuzilma — hamma narsaning asosi
Networking technologies have been in a state of constant development and improvement since the late 1960s. However, networking becomes practical and desirable and adopted into widespread use only when the proper infrastructure is in place.	Tarmoq texnologiyalari 1960-yillarning oxiridan beri doimiy ravishda rivojlanib va takomillashib kelmoqda. Biroq tarmoq texnologiyalari tegishli infratuzilma mavjud bo‘lgandagina amaliy, foydali va keng qo‘llaniladigan texnologiyaga aylanadi.
Infrastructure is the basic underlying physical structure or framework needed for the operation of a service or an enterprise. In the case of networking, infrastructure is the hardware that supports high-speed communications and data transfer. Infrastructure is millions of miles of fiber-optic cable, high-powered cell relay towers dotting the landscape, thousands of communication satellites in Earth’s orbit, high-speed routers and servers and exchange points strategically positioned along the Internet, and consumer devices capable of participating in modern networks.	Infratuzilma — xizmat yoki korxonaning ishlashi uchun zarur bo‘lgan asosiy jismoniy tuzilma yoki karkasdir. Tarmoq nuqtayi nazaridan infratuzilma yuqori tezlikdagi aloqa va ma’lumot uzatishni ta’minlaydigan apparat vositalaridir. Bunga millionlab millar optik tolali kabellar, hudud bo‘ylab joylashgan yuqori quvvatli uyali aloqa retranslyatorlari, Yer orbitasidagi minglab aloqa sun’iy yo‘ldoshlari, Internet bo‘ylab strategik joylashtirilgan yuqori tezlikdagi routerlar, serverlar va almashinuv nuqtalari hamda zamonaviy tarmoqlarda qatnasha oladigan foydalanuvchi qurilmalari kiradi.
It has taken over fifty years and billions of dollars to build the infrastructure that our modern networks rely on today. Modern networking technologies and networks evolve and expand at an exponential rate because finally there is now sufficient infrastructure to support that type of growth.	Bugungi zamonaviy tarmoqlar tayanadigan infratuzilmani yaratish uchun ellik yildan ortiq vaqt va milliardlab dollar sarflangan. Zamonaviy tarmoq texnologiyalari va tarmoqlari eksponensial tezlikda rivojlanib va kengayib bormoqda, chunki bunday o‘shishni qo‘llab-quvvatlash uchun nihoyat yetarli infratuzilma mavjud.
Connectedness is Key	Bog‘langanlik — asosiy omil
You will learn about different types of networks in this course. Even though these networks are distinct from one another, they also inter-connect with each other. Cellular networks, the telephone network, and the Internet all interconnect. Since our private home, school, or businesses networks and devices tap into these large networks, we are all interconnected.	Ushbu kursda tarmoqlarning turli turlari haqida bilib olasiz. Bu tarmoqlar bir-biridan farq qilsa-da, ular o‘zaro ham bog‘langan. Uyali aloqa tarmoqlari, telefon tarmog‘i va Internet bir-biri bilan bog‘langan. Uy, maktab yoki korxonalarimizning xususiy tarmoqlari va qurilmalari ushbu yirik tarmoqlarga ulanadi, shuning uchun barchamiz o‘zaro bog‘langanmiz.
Why Network?	Nima uchun tarmoq kerak?
Networks are in wide use today because they are practical and useful. They enable users to share many things, including: • Files – for example, Betty can create and store a file on her computer, and her fellow network user Barney can open and work with Betty’s file without having to obtain a copy of the file. • Resources – printers and scanners are commonly shared on a network. An entire office can share one or two printers, eliminating the need to purchase and attach a printer to each computer. • Internet connections – networks within a home or office allow several users to share an Internet connection.	Tarmoqlar bugungi kunda amaliy va foydali bo‘lgani uchun keng qo‘llanadi. Ular foydalanuvchilarga ko‘plab narsalarni, jumladan quyidagilarni ulashish imkonini beradi: • Fayllar — masalan, Betti o‘z kompyuterida fayl yaratib saqlashi, tarmoqdagi hamkasbi Barni esa fayl nusxasini olmasdan turib uni ochishi va ishlashi mumkin. • Resurslar — printerlar va skanerlar tarmoqda ko‘pincha umumiy foydalanishga beriladi. Butun ofis bitta yoki ikkita printerdan birgalikda foydalanishi mumkin, natijada har bir kompyuterga alohida printer sotib olish va ulash zarurati kamayadi. • Internet ulanishlari — uy yoki ofis ichidagi tarmoqlar bir nechta foydalanuvchiga bitta Internet ulanishidan birgalikda foydalanish imkonini beradi.

Study Note / O‘quv eslatmasi

- Hardware — Apparat ta’minoti / qurilmalarning jismoniy qismi
- Network — Tarmoq
- Internet — Internet

Quick Check / Tezkor nazorat

34. What is the main concept explained in this section? / Ushbu bo‘limning asosiy tushunchasi nima?
35. Which English technical terms should be remembered? / Qaysi texnik English terminlarini yodlash kerak?
36. Can you explain the concept in Uzbek without looking at the text? / Matnga qaramasdan tushunchani o‘zbekcha izohlay olasizmi?

LESSON 1 — TECHNOLOGY BASICS | SECTION 13

Manba sahifasi / Source page: 13

Lesson 1: Technology Basics

IC3 Global Standard 6 – Level 1

Network Interface Card (NIC)

In order to make an Ethernet connection between your computer and your home, school, or company wired LAN, your computer must include a network interface card (NIC). A NIC sends and receives data back and forth between your computer and the network, and includes its own Ethernet port.

NICs are manufactured to support various data transfer rates, such as 10Mbps, 100Mbps, and 1Gbps. The faster NICs are backward-compatible meaning that they can slow down to match the speed of other equipment on the network.

Note: The speed of a single network connection is only as fast as its slowest link. This means that if you have 50Mbps Internet service, a high-speed router and Cat 5e cable, but your NIC is an old 10Mbps version, your connection speed will be 10Mbps.

A NIC may be built into a device (as it is in most desktop computers and some laptops), or it can be added to a device (either internally or externally).

- You can add an internal NIC to a desktop system by opening the case and inserting the NIC into a slot on the system board.
- You can add an external NIC to a computer by connecting one to an available USB port.
- Connecting to the wired network is as easy as plugging in the cable. Most devices you connect via Ethernet automatically detect the network connection and will perform any necessary configuration (such as obtaining an IP address) automatically.



When you use a wired connection, a wired network icon  will display in the Windows taskbar.

Wireless Connections (Wi-Fi)

Wireless networking systems use their Wi-Fi capability to send and receive radio transmissions over the air instead of over cables. Wireless networking occurs in two distinct modes:

- **Ad-hoc** – systems communicate with one another directly using their Wi-Fi capability. This mode is highly insecure, and it does not provide access to the Internet.
- **Infrastructure** – systems connect to the network through a wireless router (or access point), and communicate with each other and with other systems on a wired Ethernet network through the router. This mode of communication can be secured, and it provides Internet access.



Most modern broadband routers include wireless capability, which means they can be used to create wireless networks. These are infrastructure mode networks: all wireless devices that join the network communicate with the router, not directly with one another.

Because the router is also a central connection point in the LAN, this type of network is called a Wireless LAN (WLAN). Devices connected to the WLAN can communicate (through the router) with one another and with systems that are connected to the LAN via Ethernet.

Additionally, because the router is connected to the broadband modem, the WLAN has access to the Internet.

English — O‘zbekcha

ENGLISH	O‘ZBEK
ENGLISH	O‘ZBEK
Network Interface Card (NIC)	Tarmoq interfeysi kartasi (NIC)
In order to make an Ethernet connection between your computer and your home, school, or company wired LAN, your computer must include a network interface card (NIC). A NIC sends and receives data back and forth between your computer and the network, and includes its own Ethernet port.	Kompyuteringiz bilan uy, maktab yoki kompaniyaning simli LAN tarmog‘i o‘rtasida Ethernet ulanishini tashkil qilish uchun kompyuteringizda tarmoq interfeysi kartasi (NIC) bo‘lishi kerak. NIC kompyuter va tarmoq o‘rtasida ma’lumotlarni yuboradi va qabul qiladi hamda o‘z Ethernet portiga ega.
NICs are manufactured to support various data transfer rates, such as 10Mbps, 100Mbps, and 1Gbps. The faster NICs are backward-compatible meaning that they can slow down to match the speed of other equipment on the network.	NIC‘lar 10 Mbit/s, 100 Mbit/s va 1 Gbit/s kabi turli ma’lumot uzatish tezliklarini qo‘llab-quvvatlash uchun ishlab chiqariladi. Tezroq NIC‘lar orqaga mos keladi, ya’ni ular tarmoqdagi boshqa uskunalarning tezligiga moslashish uchun o‘z tezligini pasaytiradi.
Note: The speed of a single network connection is only as fast as its slowest link. This means that if you have 50Mbps Internet service, a high-speed router and Cat 5e cable, but your NIC is an old 10Mbps version, your connection speed will be 10Mbps.	Eslatma: bitta tarmoq ulanishining tezligi uning eng sekin bo‘g‘ini tezligidan oshmaydi. Masalan, sizda 50 Mbit/s Internet xizmati, yuqori tezlikdagi router va Cat 5e kabel bo‘lsa, ammo NIC‘ingiz eski 10 Mbit/s versiya bo‘lsa, ulanish tezligi 10 Mbit/s bo‘ladi.
A NIC may be built into a device (as it is in most desktop computers and some laptops), or it can be added to a device (either internally or externally). • You can add an internal NIC to a desktop system by opening the case and inserting the NIC into a slot on the system board. • You can add an external NIC to a computer by connecting one to an available USB port.	NIC qurilmaning ichiga o‘rnatilgan bo‘lishi mumkin (ko‘pchilik stol kompyuterlari va ayrim noutbuklarda shunday) yoki qurilmaga ichki yoxud tashqi tarzda qo‘shilishi mumkin. • Stol kompyuteriga ichki NIC qo‘shish uchun korpusni ochib, NIC‘ni tizim platasidagi slotga o‘rnatish mumkin. • Kompyuterga tashqi NIC‘ni mavjud USB portga ulash orqali qo‘shish mumkin.
Connecting to the wired network is as easy as plugging in the cable. Most devices you connect via Ethernet automatically detect the network connection and will perform any necessary configuration (such as obtaining an IP address) automatically. When you use a wired connection, a wired network icon will display in the Windows taskbar.	Simli tarmoqqa ulanish kabelni ulash kabi oson. Ethernet orqali ulangan ko‘pchilik qurilmalar tarmoq ulanishini avtomatik aniqlaydi va zarur sozlamalarni, masalan IP manzilini olishni, avtomatik bajaradi. Simli ulanishdan foydalanganda Windows vazifalar panelida simli tarmoq belgisi ko‘rsatiladi.
Wireless Connections (Wi-Fi)	Simsiz ulanishlar (Wi-Fi)
Wireless networking systems use their Wi-Fi capability to send and receive radio transmissions over the air instead of over cables. Wireless networking occurs in two distinct modes: • Ad-hoc – systems communicate with one another directly using their Wi-Fi capability. This mode is highly insecure, and it does not provide access to the Internet. • Infrastructure – systems connect to the network through a wireless router (or access point), and communicate with each other and with other systems on a wired Ethernet network through the router. This mode of communication can be secured, and it provides Internet access.	Simsiz tarmoq tizimlari kabel o‘rniga havoda radio uzatmalarini yuborish va qabul qilish uchun Wi-Fi imkoniyatidan foydalanadi. Simsiz tarmoq ikki asosiy rejimda ishlaydi: • Ad-hoc — tizimlar Wi-Fi imkoniyatidan foydalanib bir-biri bilan bevosita aloqa qiladi. Bu rejim xavfsizligi past bo‘lib, Internetga kirishni ta’minlamaydi. • Infrastructure — tizimlar simsiz router (yoki access point) orqali tarmoqqa ulanadi va router orqali bir-biri hamda simli Ethernet tarmog‘idagi boshqa tizimlar bilan aloqa qiladi. Bu aloqa rejimini himoyalash mumkin va u Internetga kirishni ta’minlaydi.
Most modern broadband routers include wireless capability, which means they can be used to create wireless networks. These are infrastructure mode networks: all wireless devices that join the network communicate with the router, not directly with one another.	Ko‘pchilik zamonaviy keng polosali routerlarda simsiz aloqa imkoniyati mavjud, ya’ni ulardan simsiz tarmoqlar yaratish mumkin. Bular infrastructure rejimidagi tarmoqlardir: tarmoqqa qo‘shilgan barcha simsiz qurilmalar bir-biri bilan bevosita emas, router orqali aloqa qiladi.
Because the router is also a central connection point in the LAN, this type of network is called a Wireless LAN (WLAN). Devices connected to the WLAN can communicate (through the router) with one another and with systems that are connected to the LAN via Ethernet.	Router LAN ichidagi markaziy ulanish nuqtasi ham bo‘lgani uchun bunday tarmoq Wireless LAN (WLAN), ya’ni simsiz LAN deb ataladi. WLAN‘ga ulangan qurilmalar router orqali bir-biri bilan va Ethernet orqali LAN‘ga ulangan tizimlar bilan aloqa qilishi mumkin.
Additionally, because the router is connected to the broadband modem, the WLAN has access to the Internet.	Bundan tashqari, router keng polosali modemga ulanganligi sababli WLAN Internetga kirish imkoniga ega bo‘ladi.

Study Note / O‘quv eslatmasi

- Network — Tarmoq

- Internet — Internet

Quick Check / Tezkor nazorat

37. What is the main concept explained in this section? / Ushbu bo‘limning asosiy tushunchasi nima?
38. Which English technical terms should be remembered? / Qaysi texnik English terminlarini yodlash kerak?
39. Can you explain the concept in Uzbek without looking at the text? / Matnga qaramasdan tushunchani o‘zbekcha izohlay olasizmi?

LESSON 1 — TECHNOLOGY BASICS | SECTION 14

Manba sahifasi / Source page: 14

Lesson 1: Technology Basics

IC3 Global Standard 6 – Level 1

- If you do call your ISP, be prepared to provide information to the technical specialist such as your current IP address, and your connection speed. Remember that you view your network connections in the Network and Sharing Center, and that you can click your connection to view necessary details.
- If you have trouble connecting to your Wi-Fi network, check the following:
 - Is your wireless adapter turned on? Sometimes, wireless adapters are turned off by a device in order to save battery power.
 - Has the Wi-Fi network password changed? Network administrators sometimes change the Wi-Fi password if they suspect that the previous one has been compromised or if they conduct scheduled password changes.
 - Are you close enough to the Wi-Fi hotspot? WLAN devices have a limited range, especially if they are using older WLAN standards such as 802.11b or 802.11g.

Trouble shooting is a process that takes practice - the more you do it, the better you will become.

Lesson Summary

In this lesson, you explored the following concepts:

1	Technology Basics – Level 1	
	1.1	Access and navigate between digital environments
	1.1.1	Recognize operating system terms and concepts
	1.1.2	Explain basic functions of web browsers
	1.1.3	Explain processes and requirements for accessing digital
	1.1.4	Explain methods of navigating between digital environments
	1.2	Identify digital devices and connections
	1.2.1	Identify input devices
	1.2.2	Identify output devices
	1.2.3	Identify cables, connectors, and connections
	1.3	Explain fundamental software concepts
	1.3.1	Explain basic software application concepts
	1.3.2	Compare and contrast proprietary and open-source software
	1.3.3	Describe processes for installing software from online sources
	1.4	Explain fundamental hardware concepts
	1.4.1	Describe concepts related to computing devices
	1.4.2	Describe concepts related to memory
	1.4.3	Describe concepts related to data storage
	1.5	Explain fundamental operating system concepts
	1.5.1	Compare and contrast features of mobile device operating systems
	1.5.2	Compare and contrast features of computer operating systems

English — O‘zbekcha

ENGLISH	O‘ZBEK
ENGLISH	O‘ZBEK
If you do call your ISP, be prepared to provide information to the technical specialist such as your current IP address, and your connection speed. Remember that you view your network connections in the Network and Sharing Center, and that you can click your connection to view necessary details.	Agar ISP'ingizga qo'ng'iroq qilsangiz, texnik mutaxassisga joriy IP manzilingiz va ulanish tezligingiz kabi ma'lumotlarni taqdim etishga tayyor bo'ling. Tarmoq ulanishlarini Network and Sharing Center bo'limida ko'rishingiz va kerakli tafsilotlarni ko'rish uchun ulanish nomini bosishingiz mumkinligini unutmang.
If you have trouble connecting to your Wi-Fi network, check the following: – Is your wireless adapter turned on? Sometimes, wireless adapters are turned off by a device in order to save battery power. – Has the Wi-Fi network password changed? Network administrators sometimes change the Wi-Fi password if they suspect that the previous one has been compromised or if they conduct scheduled password changes. – Are you close enough to the Wi-Fi hotspot? WLAN devices have a limited range, especially if they are using older WLAN standards such as 802.11b or 802.11g.	Wi-Fi tarmog'iga ulanishda muammo yuzaga kelsa, quyidagilarni tekshiring: – Simsiz adapter yoqilganmi? Ba'zan batareya quvvatini tejash uchun qurilma simsiz adapterni o'chirib qo'yadi. – Wi-Fi tarmog'i paroli o'zgarganmi? Tarmoq administratorlari avvalgi parol buzilgan deb gumon qilsalar yoki rejalashtirilgan parol almashtirishni amalga oshirsalar, Wi-Fi parolini o'zgartirishlari mumkin. – Wi-Fi hotspotiga yetarlicha yaqinmisiz? WLAN qurilmalarining qamrov masofasi cheklangan, ayniqsa 802.11b yoki 802.11g kabi eski WLAN standartlaridan foydalanilganda.
Troubleshooting is a process that takes practice - the more you do it, the better you will become.	Muammolarni bartaraf etish (troubleshooting) mashq qilishni talab qiladigan jarayondir — qancha ko'p bajarsangiz, shuncha yaxshi o'zlashtirasiz.
Lesson Summary In this lesson, you explored the following concepts: 1 Technology Basics – Level 1 1.1 Access and navigate between digital environments 1.1.1 Recognize operating system terms and concepts 1.1.2 Explain basic functions of web browsers 1.1.3 Explain processes and requirements for accessing digital 1.1.4 Explain methods of navigating between digital environments 1.2 Identify digital devices and connections 1.2.1 Identify input devices 1.2.2 Identify output devices 1.2.3 Identify cables, connectors, and connections 1.3 Explain fundamental software concepts 1.3.1 Explain basic software application concepts 1.3.2 Compare and contrast proprietary and open-source software 1.3.3 Describe processes for installing software from online sources 1.4 Explain fundamental hardware concepts 1.4.1 Describe concepts related to computing devices 1.4.2 Describe concepts related to memory 1.4.3 Describe concepts related to data storage 1.5 Explain fundamental operating system concepts 1.5.1 Compare and contrast features of mobile device operating systems 1.5.2 Compare and contrast features of computer operating systems	Dars xulosasi Ushbu darsda quyidagi tushunchalarni o'rgandingiz: 1 Texnologiya asoslari — 1-daraja 1.1 Raqamli muhitlarga kirish va ular orasida navigatsiya qilish 1.1.1 Operatsion tizim atamalari va tushunchalarini tanib olish 1.1.2 Veb-brauzerlarning asosiy funksiyalarini tushuntirish 1.1.3 Raqamli muhitlarga kirish jarayonlari va talablarini tushuntirish 1.1.4 Raqamli muhitlar orasida navigatsiya qilish usullarini tushuntirish 1.2 Raqamli qurilmalar va ulanishlarni aniqlash 1.2.1 Kiritish qurilmalarini aniqlash 1.2.2 Chiqarish qurilmalarini aniqlash 1.2.3 Kabellar, ulagichlar va ulanishlarni aniqlash 1.3 Dasturiy ta'minotning asosiy tushunchalarini tushuntirish 1.3.1 Amaliy dasturiy ta'minotning asosiy tushunchalarini tushuntirish 1.3.2 Xususiy va ochiq kodli dasturiy ta'minotni taqqoslash 1.3.3 Dasturiy ta'minotni onlayn manbalardan o'rnatish jarayonlarini tavsiflash 1.4 Apparat ta'minotining asosiy tushunchalarini tushuntirish 1.4.1 Hisoblash qurilmalariga oid tushunchalarni tavsiflash 1.4.2 Xotiraga oid tushunchalarni tavsiflash 1.4.3 Ma'lumotlarni saqlashga oid tushunchalarni tavsiflash 1.5 Operatsion tizimning asosiy tushunchalarini tushuntirish 1.5.1 Mobil qurilma operatsion tizimlari xususiyatlarini taqqoslash 1.5.2 Kompyuter operatsion tizimlari xususiyatlarini taqqoslash

Study Note / O'quv eslatmasi

- Hardware — Apparat ta'minoti / qurilmalarning jismoniy qismi
- Software — Dasturiy ta'minot
- Input device — Kiritish qurilmasi
- Output device — Chiqarish qurilmasi
- Network — Tarmoq

- Browser — Brauzer
- Open-source — Ochiq kodli
- Proprietary — Xususiy / yopiq litsenziyali

Quick Check / Tezkor nazorat

40. What is the main concept explained in this section? / Ushbu bo‘limning asosiy tushunchasi nima?
41. Which English technical terms should be remembered? / Qaysi texnik English terminlarini yodlash kerak?
42. Can you explain the concept in Uzbek without looking at the text? / Matnga qaramasdan tushunchani o‘zbekcha izohlay olasizmi?

LESSON 1 — TECHNOLOGY BASICS | SECTION 15

Manba sahifasi / Source page: 15

Lesson 1: Technology Basics

IC3 Global Standard 6 – Level 1

23. If you are uncertain as to whether your device has a working Internet connection, which one of the following actions can you take to quickly check it?
- Check the connections on the network router to ensure they are connected.
 - Contact your ISP to request them to verify your connection.
 - Check your email send/receive option.
 - Open your browser to see if you are able to access a web site.
24. Which of the following statements about Bluetooth technology are TRUE? Select the four options that apply.
- Bluetooth technology permits enabled devices to communicate directly with each other.
 - Bluetooth only needs to be enabled on one of the two devices.
 - In order to be paired, both Bluetooth devices must be in discovery mode.
 - Bluetooth accessories include headsets, mice, keyboards, and automobile media devices.
 - Bluetooth accessories sometimes require a PIN to be entered to ensure secure communication.
 - Bluetooth devices must be paired in order to communicate.
 - Once paired, Bluetooth devices automatically connect when powered up and within range.
25. Computer networks and the Internet use both private and public IP addresses to know where information is coming from and where it should be sent.
- From the statements below, select those that are TRUE for *Private* IP addresses. Select the four options that apply.
- IP address obtained from a DHCP server.
 - Reserved for internal use behind a router.
 - Used by DNS to locate data destinations
 - Identifies each device on the local network.
 - Translates to a common URL
 - Static IP address and does not change.
 - 198.162.x.1 identifies the default gateway.
36. Computer networks and the Internet use both private and public IP addresses to know where information is coming from and where it should be sent.
- From the statements below, select those that are TRUE for *Public* IP addresses. Select the three options that apply.
- IP address obtained from a DHCP server.
 - Reserved for internal use behind a router.
 - Used by DNS to locate data destinations
 - Identifies each device on the local network.
 - Translates to a common URL
 - Static IP address and does not change.
 - 198.162.x.1 identifies the default gateway.

English — O'zbekcha

ENGLISH	O'ZBEK
ENGLISH	O'ZBEK
23. If you are uncertain as to whether your device has a working Internet connection, which one of the following actions can you take to quickly check it? a. Check the connections on the network router to ensure they are connected. b. Contact your ISP to request them to verify your connection. c. Check your email send/receive option. d. Open your browser to see if you are able to access a web site.	23. Qurilmangizda Internet ulanishi ishlayotganiga ishonchingiz komil bo'lmasa, uni tezda tekshirish uchun quyidagi amallardan qaysi birini bajarishingiz mumkin? a. Tarmoq routeridagi ulanishlar ulanganini tekshirish. b. ISP'ga murojaat qilib, ulanishni tekshirishni so'rash. c. Elektron pochtaning yuborish/qabul qilish funksiyasini tekshirish. d. Veb-saytga kira olishingizni tekshirish uchun brauzerni ochish.
24. Which of the following statements about Bluetooth technology are TRUE? Select the four options that apply. a. Bluetooth technology permits enabled devices to communicate directly with each other. b. Bluetooth only needs to be enabled on one of the two devices. c. In order to be paired, both Bluetooth devices must be in discovery mode. d. Bluetooth accessories include headsets, mice, keyboards, and automobile media devices. e. Bluetooth accessories sometimes require a PIN to be entered to ensure secure communication. f. Bluetooth devices must be paired in order to communicate. g. Once paired, Bluetooth devices automatically connect when powered up and within range.	24. Bluetooth texnologiyasi haqidagi quyidagi fikrlardan qaysilari TO'G'RI? Mos keladigan to'rtta variantni tanlang. a. Bluetooth texnologiyasi yoqilgan qurilmalarga bir-biri bilan bevosita aloqa qilish imkonini beradi. b. Bluetooth ikki qurilmadan faqat bittasida yoqilgan bo'lishi kifoya. c. Juftlashtirish uchun ikkala Bluetooth qurilmasi ham discovery rejimida bo'lishi kerak. d. Bluetooth aksessuarlariga garnituralar, sichqonchalar, klaviaturalar va avtomobil media qurilmalari kiradi. e. Bluetooth aksessuarlari xavfsiz aloqa uchun ba'zan PIN kiritishni talab qiladi. f. Bluetooth qurilmalari aloqa qilish uchun juftlashtirilishi kerak. g. Bir marta juftlashtirilgach, Bluetooth qurilmalari yoqilganda va bir-birining qamrov hududida bo'lsa, avtomatik ulanadi.
25. Computer networks and the Internet use both private and public IP addresses to know where information is coming from and where it should be sent. From the statements below, select those that are TRUE for Private IP addresses. Select the four options that apply. a. IP address obtained from a DHCP server. b. Reserved for internal use behind a router. c. Used by DNS to locate data destinations. d. Identifies each device on the local network. e. Translates to a common URL. f. Static IP address and does not change. g. 198.162.x.1 identifies the default gateway.	25. Kompyuter tarmoqlari va Internet axborot qayerdan kelayotganini va qayerga yuborilishi kerakligini aniqlash uchun xususiy va ommaviy IP manzillardan foydalanadi. Quyidagi fikrlardan qaysilari xususiy IP manzillarga (Private IP addresses) tegishli va TO'G'RI? Mos keladigan to'rtta variantni tanlang. a. DHCP serveridan olinadigan IP manzil. b. Router ortida ichki foydalanish uchun rezerv qilingan. c. Ma'lumot manzillarini aniqlash uchun DNS tomonidan ishlatiladi. d. Mahalliy tarmoqdagi har bir qurilmani aniqlaydi. e. Umumiy URL manzilga tarjima qilinadi. f. Statik IP manzil bo'lib, o'zgarmaydi. g. 198.162.x.1 standart shlyuzni aniqlaydi.
36. Computer networks and the Internet use both private and public IP addresses to know where information is coming from and where it should be sent. From the statements below, select those that are TRUE for Public IP addresses. Select the three options that apply. a. IP address obtained from a DHCP server. b. Reserved for internal use behind a router. c. Used by DNS to locate data destinations. d. Identifies each device on the local network. e. Translates to a common URL. f. Static IP address and does not change. g. 198.162.x.1 identifies the default gateway.	36. Kompyuter tarmoqlari va Internet axborot qayerdan kelayotganini va qayerga yuborilishi kerakligini aniqlash uchun xususiy va ommaviy IP manzillardan foydalanadi. Quyidagi fikrlardan qaysilari ommaviy IP manzillarga (Public IP addresses) tegishli va TO'G'RI? Mos keladigan uchta variantni tanlang. a. DHCP serveridan olinadigan IP manzil. b. Router ortida ichki foydalanish uchun rezerv qilingan. c. Ma'lumot manzillarini aniqlash uchun DNS tomonidan ishlatiladi. d. Mahalliy tarmoqdagi har bir qurilmani aniqlaydi. e. Umumiy URL manzilga tarjima qilinadi. f. Statik IP manzil bo'lib, o'zgarmaydi. g. 198.162.x.1 standart shlyuzni aniqlaydi.

Study Note / O'quv eslatmasi

- Network — Tarmoq
- Internet — Internet
- Browser — Brauzer

- Bluetooth — Bluetooth simsiz aloqa texnologiyasi

Quick Check / Tezkor nazorat

43. What is the main concept explained in this section? / Ushbu bo‘limning asosiy tushunchasi nima?
44. Which English technical terms should be remembered? / Qaysi texnik English terminlarini yodlash kerak?
45. Can you explain the concept in Uzbek without looking at the text? / Matnga qaramasdan tushunchani o‘zbekcha izohlay olasizmi?

IC3 Exam Preparation / Imtihonga tayyorgarlik

Eslatma: Quyidagi checklist manbadagi Lesson 1 objective'larini qayta ko'rib chiqish uchun tuzilgan. U manbadagi tushunchalarni tartibga soladi va mustaqil takrorlashga yordam beradi.

- Operating system (OS) nima va u hardware hamda application programlar bilan qanday bog'lanadi?
- Desktop, mobile va embedded operating system tushunchalarini farqlang.
- Browsing asosiy vazifalarini va address bar tushunchasini tushuntiring.
- Input va output device'larni ajrating.
- Keyboard, mouse, scanner, microphone, monitor, printer va speakers kabi qurilmalarni funksiyasi bo'yicha tasniflang.
- USB, Bluetooth va boshqa ulanishlarning vazifasini eslab qoling.
- Locally-installed software va cloud-based software tushunchalarini solishtiring.
- Proprietary va open-source software farqini tushuntiring.
- Software o'rnatish va EULA/registration tushunchalarini ko'rib chiqing.
- Desktop va laptop computer xususiyatlarini solishtiring.
- Memory va data storage tushunchalarini ajrating.
- Network, infrastructure va internetworking tushunchalarini izohlang.
- Network orqali files, resources va Internet connection kabi resurslarni ulashish g'oyasini tushuning.

Bilingual Glossary / Ikki tilli lug'at

English	O'zbekcha
address bar	manzil satri
browser	brauzer
computer network	kompyuter tarmog'i
desktop computer	stol kompyuteri
laptop	noutbuk
keyboard	klaviatura
mouse	sichqoncha
monitor	monitor
printer	printer
scanner	skaner
microphone	mikrofon
speaker	karnay/dinamik
touchscreen	sensorli ekran
USB hub	USB konsentratori
wireless	simsiz
server	server
router	marshrutizator
network interface card	tarmoq interfeys kartasi
file	fayl
folder	papka
download	yuklab olish
upload	yuklash
application	amaliy dastur
operating system	operatsion tizim
memory	xotira
storage	ma'lumot saqlash xotirasi

Revision Planner / Takrorlash rejasi

1-kun: Operating System + Browser

2-kun: Input/Output Devices

3-kun: Cables, Connectors + Bluetooth

4-kun: Software + Application Programs

5-kun: Hardware + Memory + Storage

6-kun: Operating Systems

7-kun: Networking + umumiy takrorlash

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